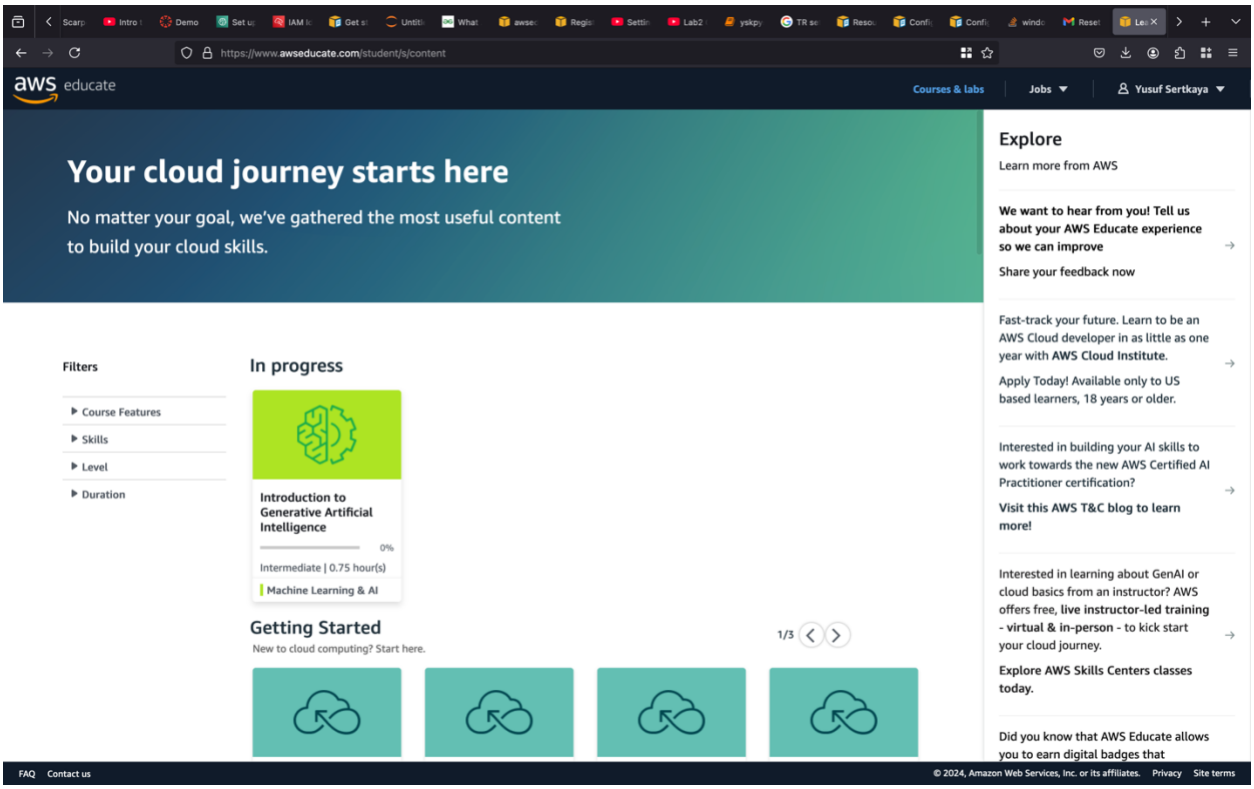
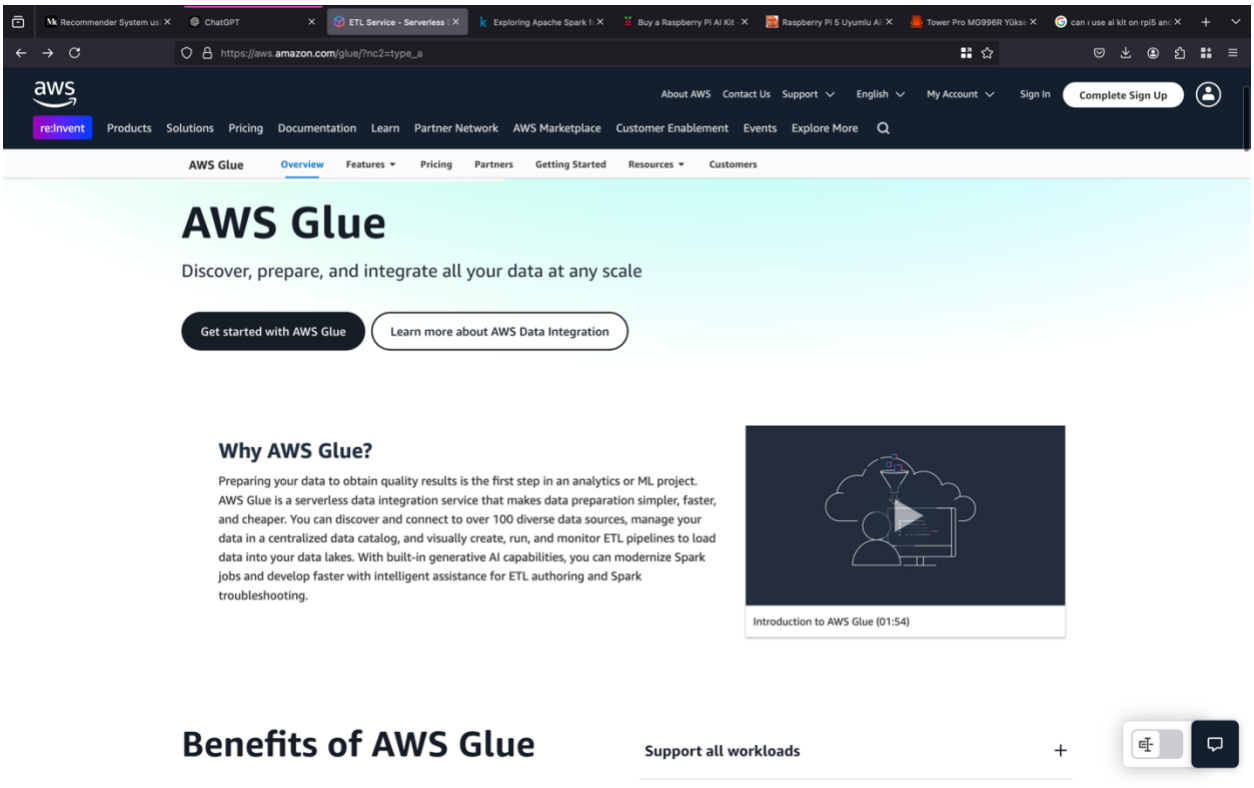
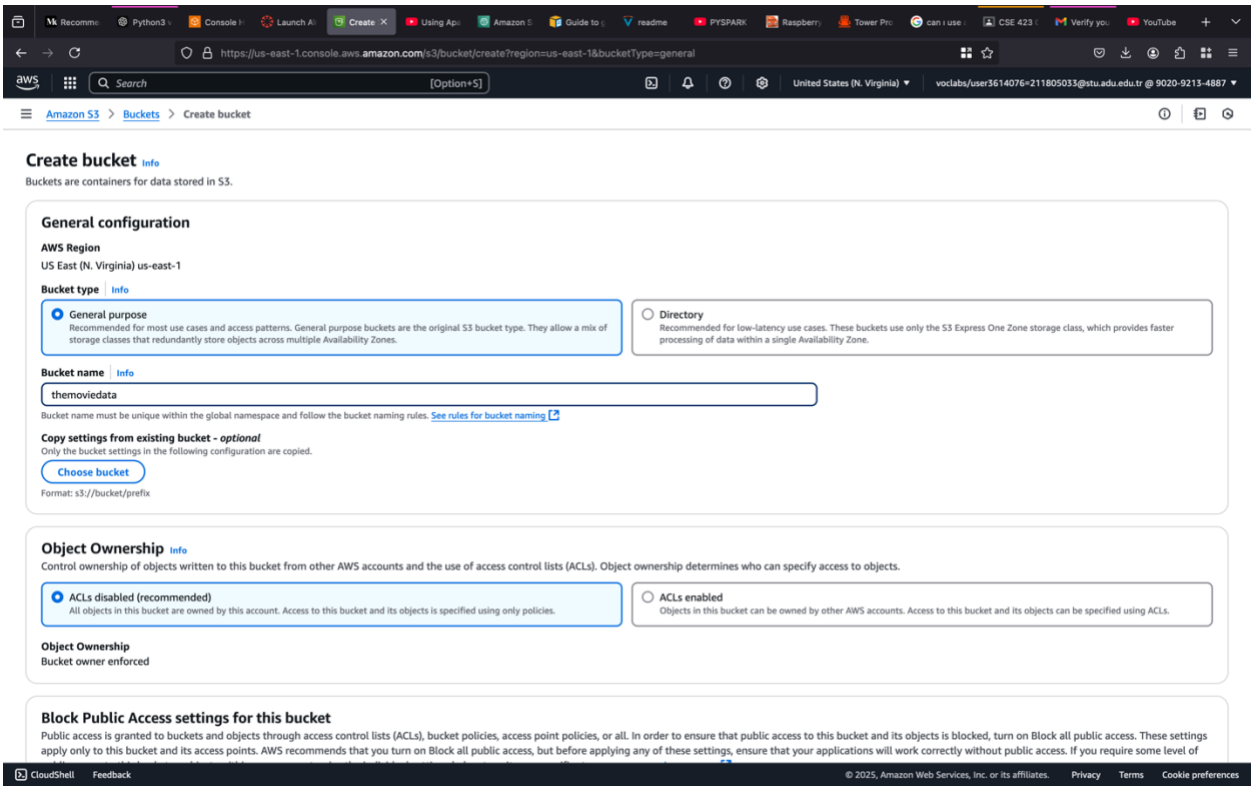


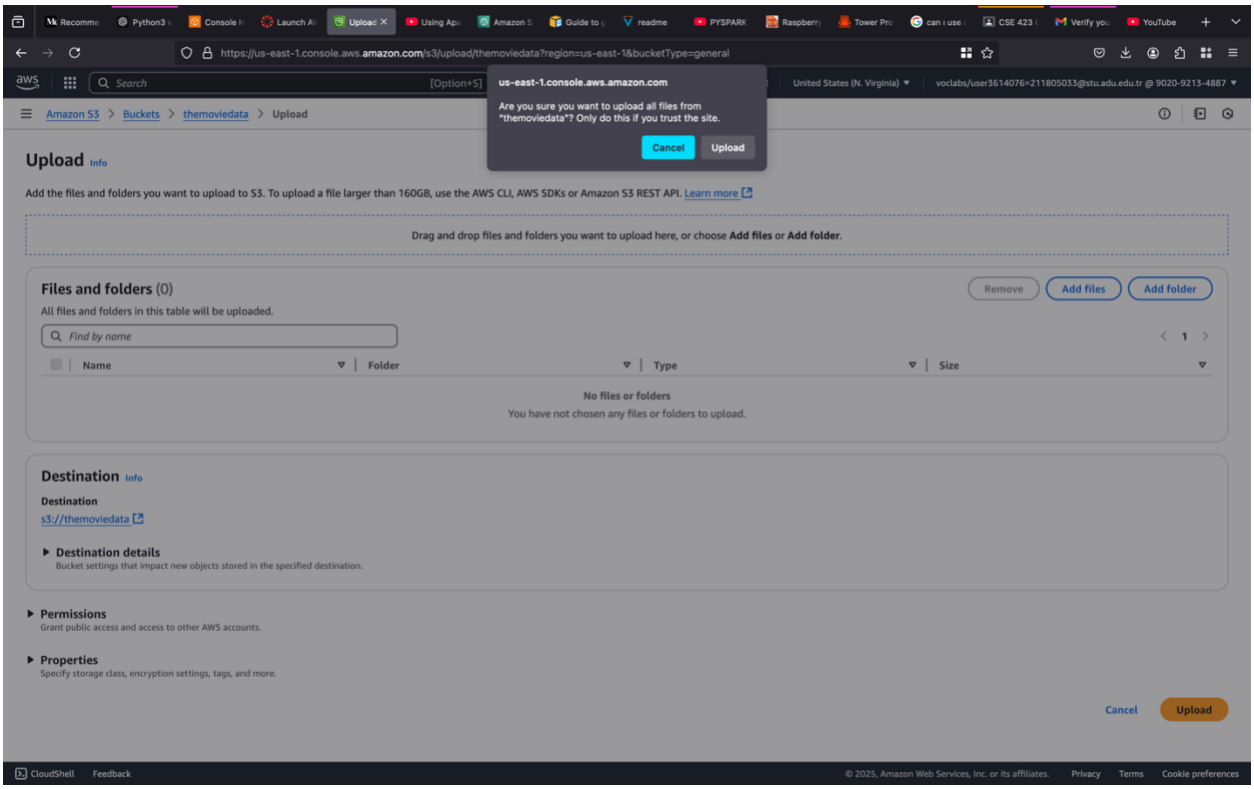
I watched some tutorials on aws educate.



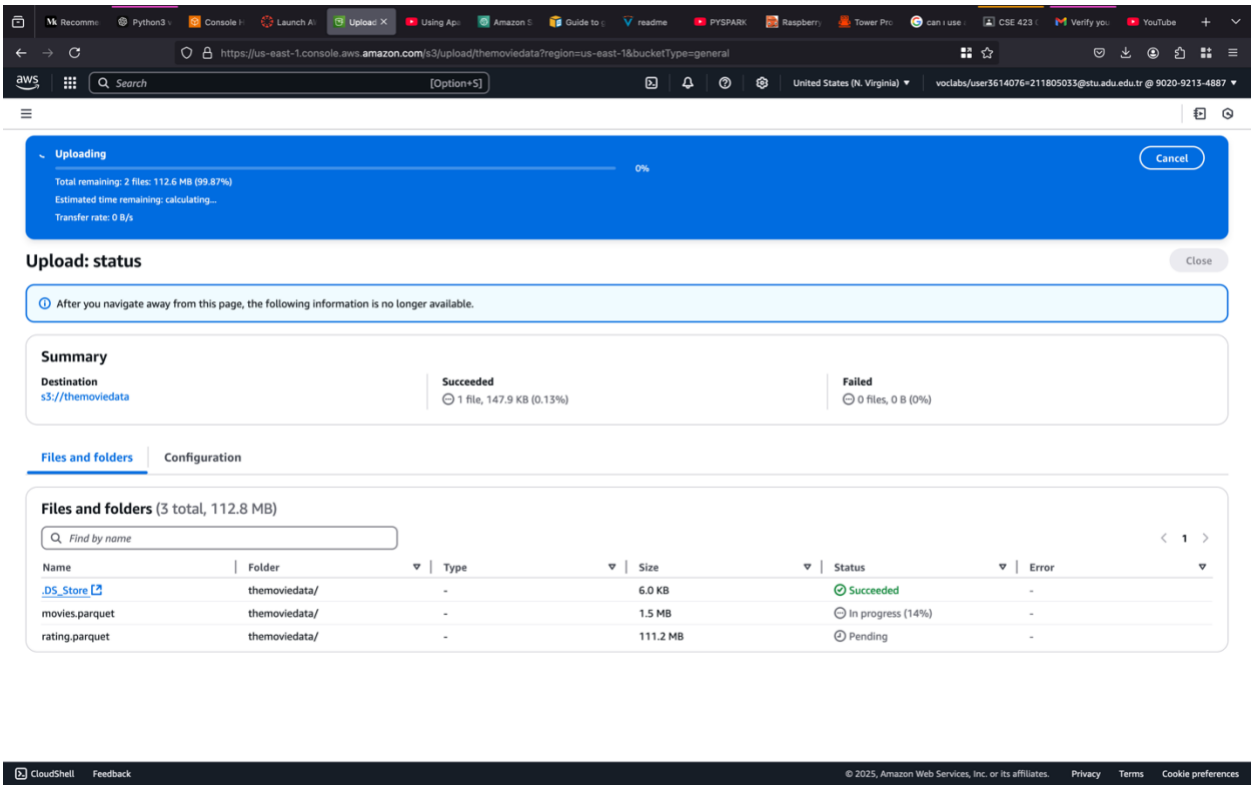
The primary goal of this project was to develop a machine learning pipeline within AWS, with a particular focus on leveraging the PySpark framework for big data processing. Additionally, I aimed to explore and gain hands-on experience with various AWS services. To achieve this, I conducted thorough research, followed hands-on tutorials, and reviewed official AWS documentation. As a starting point, I began working with AWS Glue, which serves as a core component for data integration and ETL (Extract, Transform, Load) processes within the AWS ecosystem.



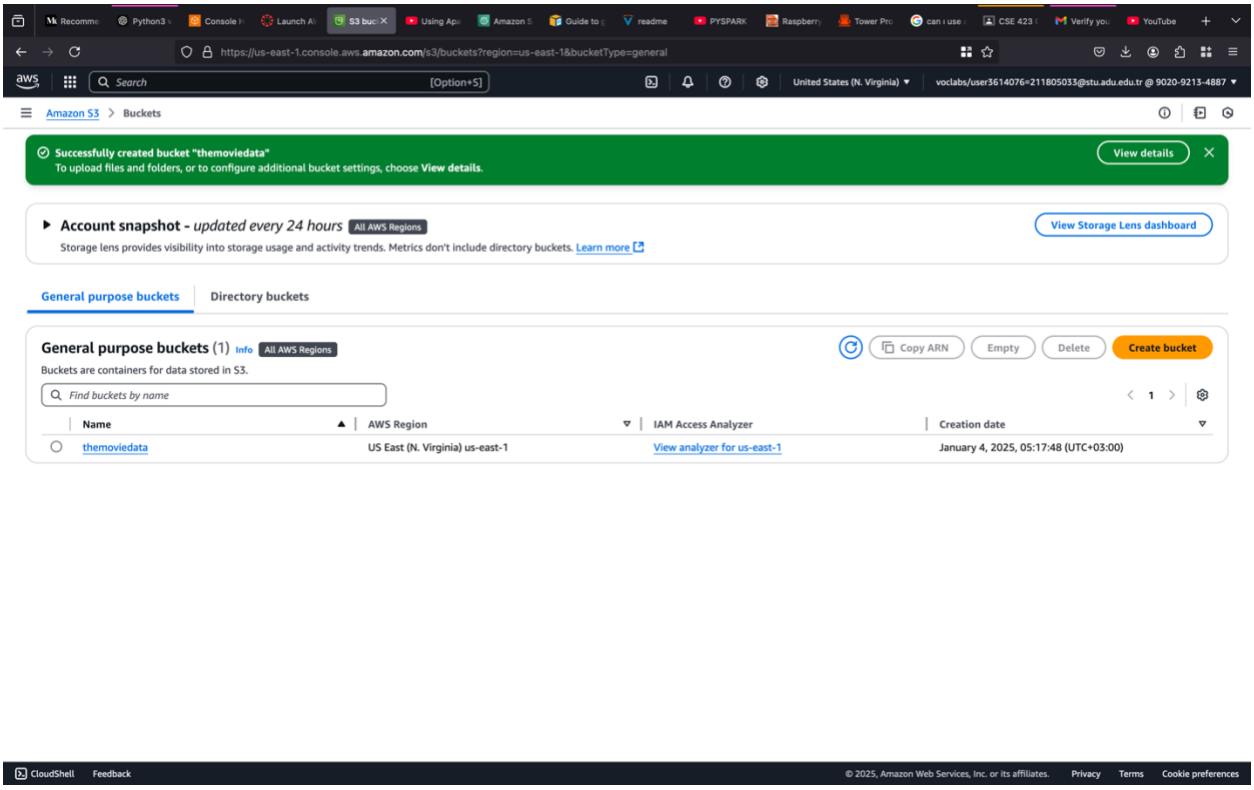
To facilitate data storage and processing, I created an Amazon S3 bucket as a secure and scalable repository for the movie dataset I sourced from Kaggle. This dataset serves as the foundation for the project, enabling efficient data access and management for subsequent stages of the machine learning pipeline. Using S3 ensures high availability and durability, making it an ideal choice for big data workflows in AWS.



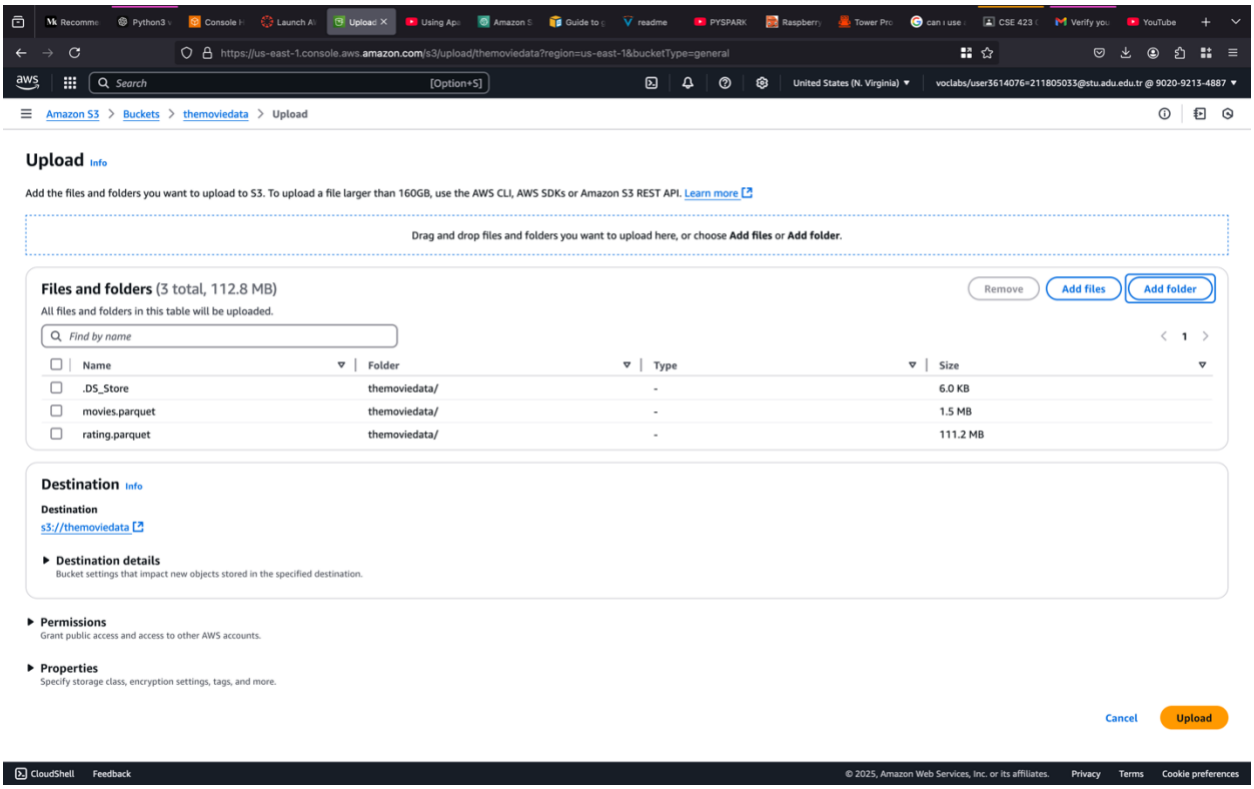
I show the proccess when I load the data to the s3 bucket.



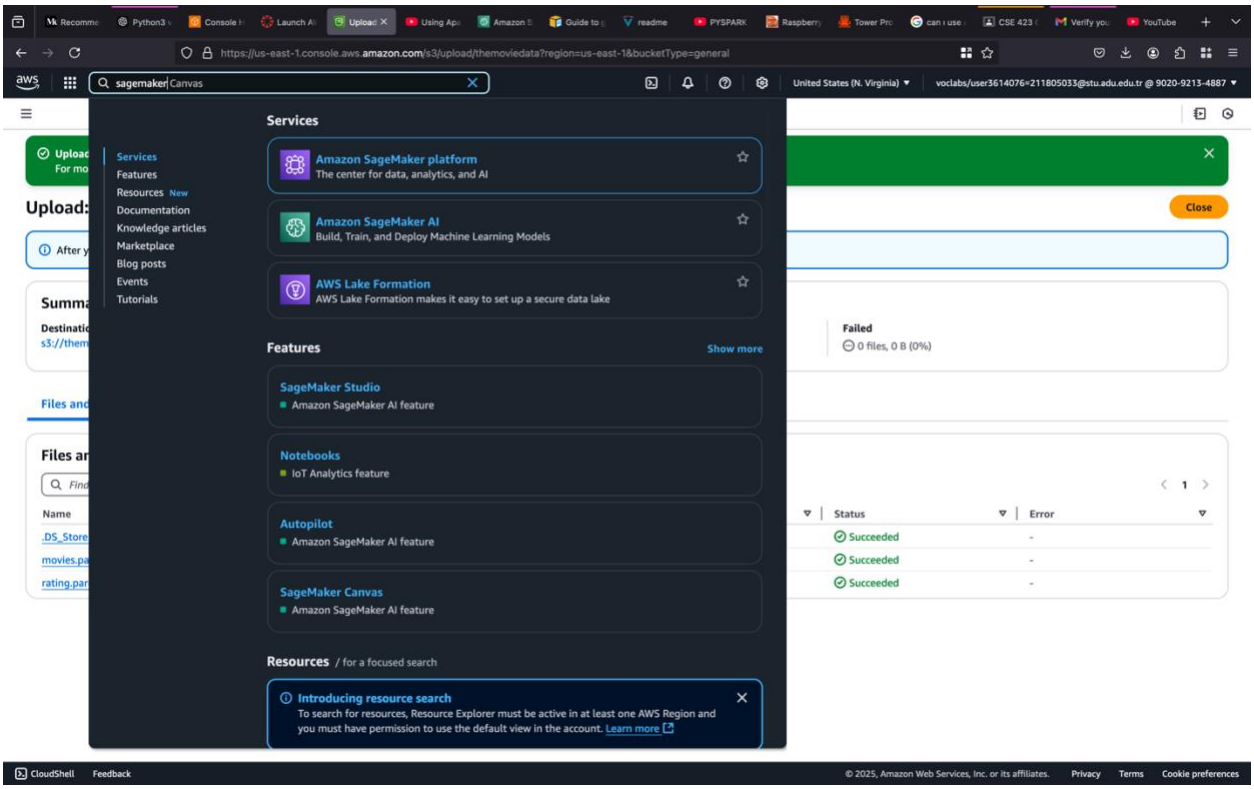
Additanioly I choose to use parquet instead of csv because it is more efficent to work with parquet instead of csv or other formats.



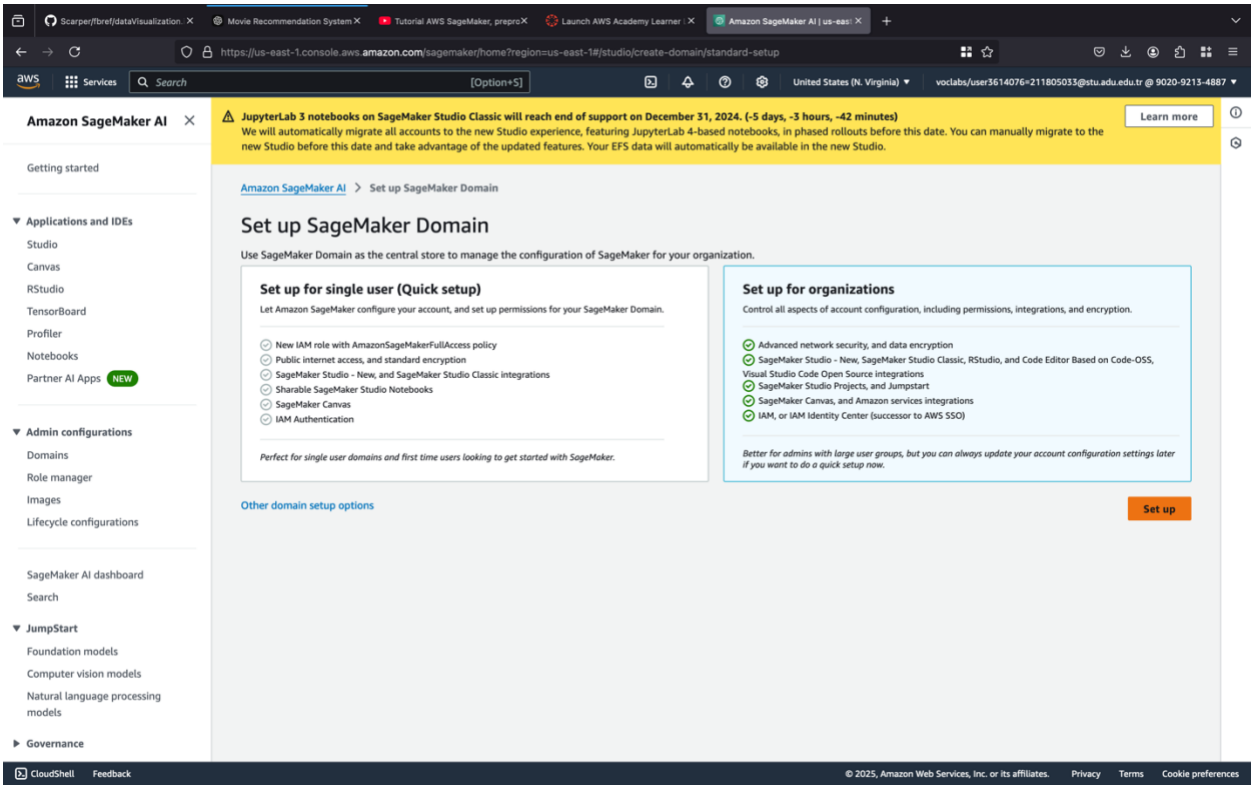
Finnaly succesfully created and stored the data.



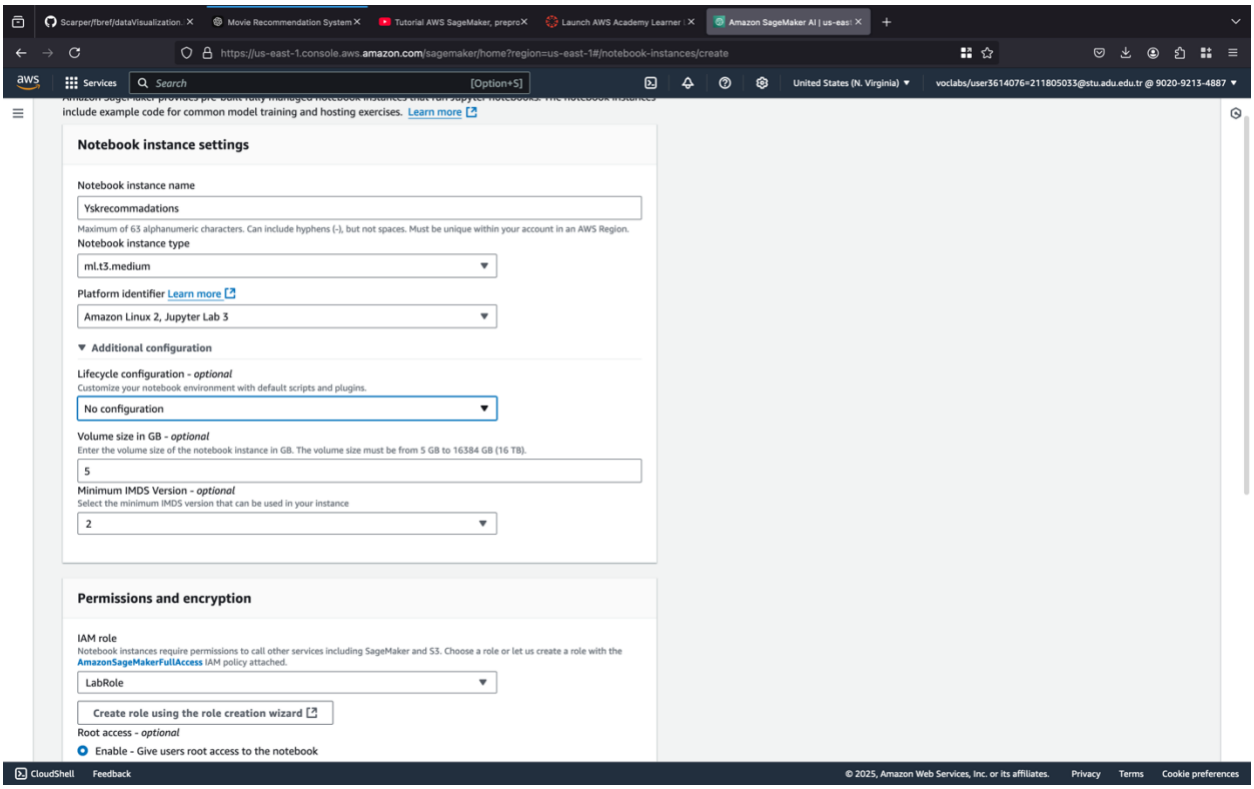
There is two part of this data one focused on movies other one on ratings and users.



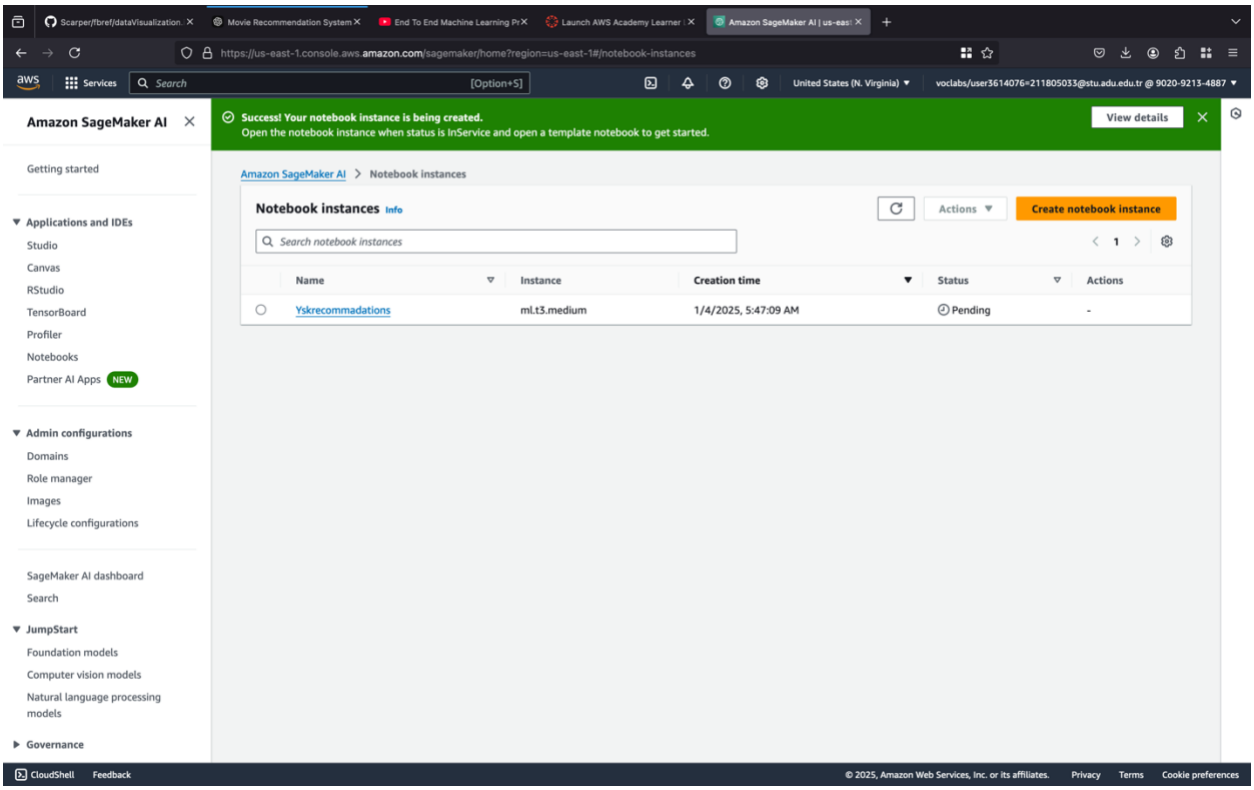
As part of my exploration of AWS services, I researched AWS SageMaker, a fully managed service designed to build, train, and deploy machine learning models at scale. To deepen my understanding, I engaged in hands-on activities, experimenting with its features and functionalities. This practical experience allowed me to familiarize myself with SageMaker's capabilities, such as creating Jupyter notebooks, training models using built-in algorithms, and deploying endpoints for real-time predictions.



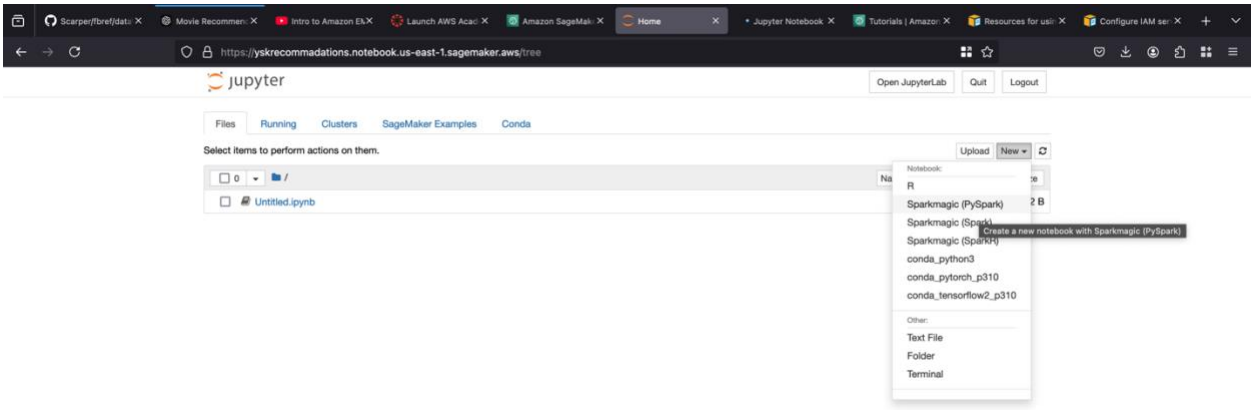
I was not able to choose for single user I had to choose a set up for organizations so choose that and move the next step.



I configured notebook instance settings.

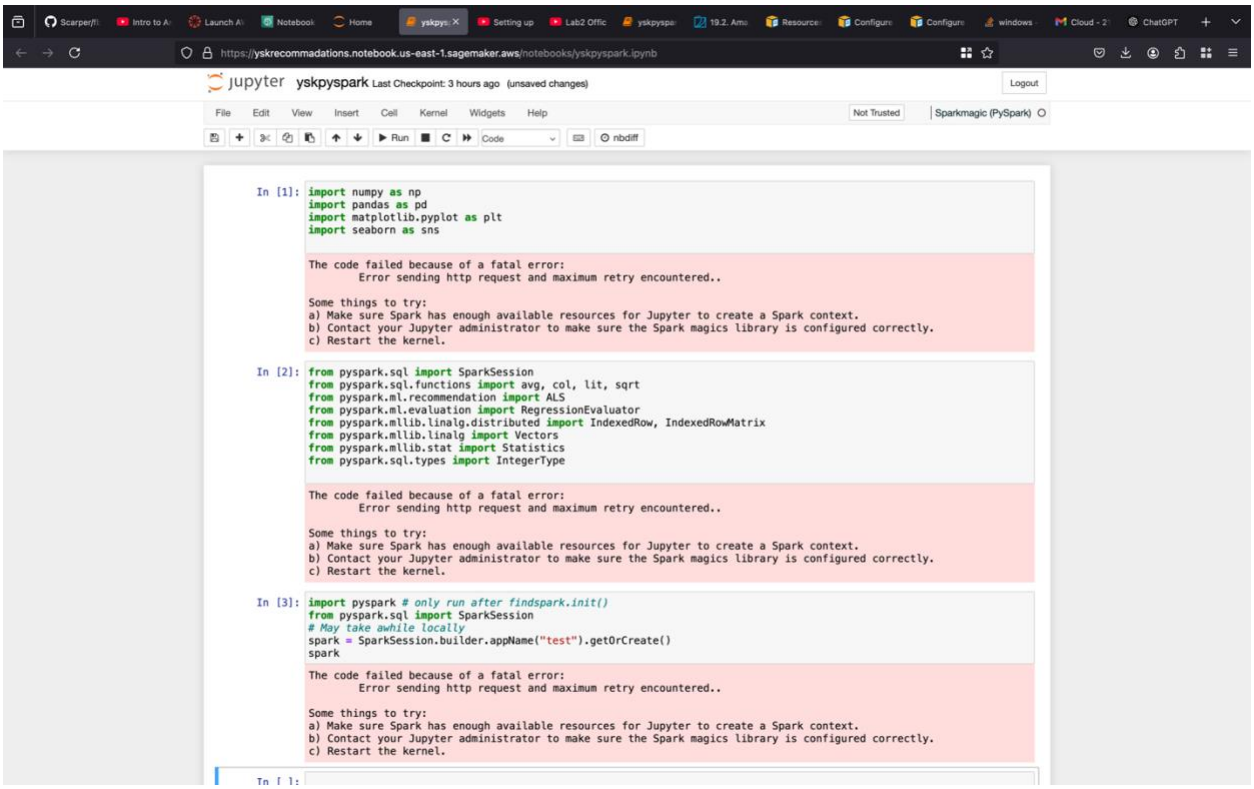


I succesfully created the notebook.

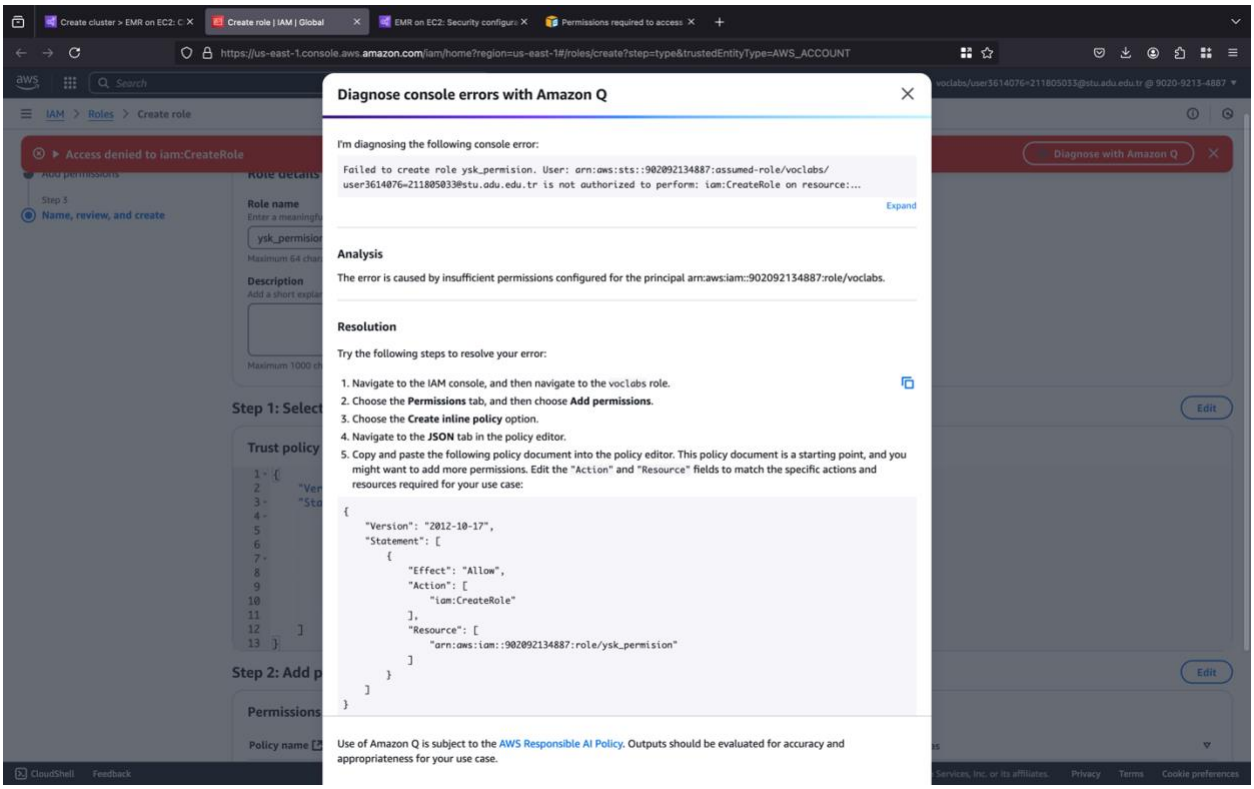


https://yskrecommadations.notebook.us-east-1.sagemaker.aws/tree#

I used Sparkmagis (PySpark) because I will use that framework.

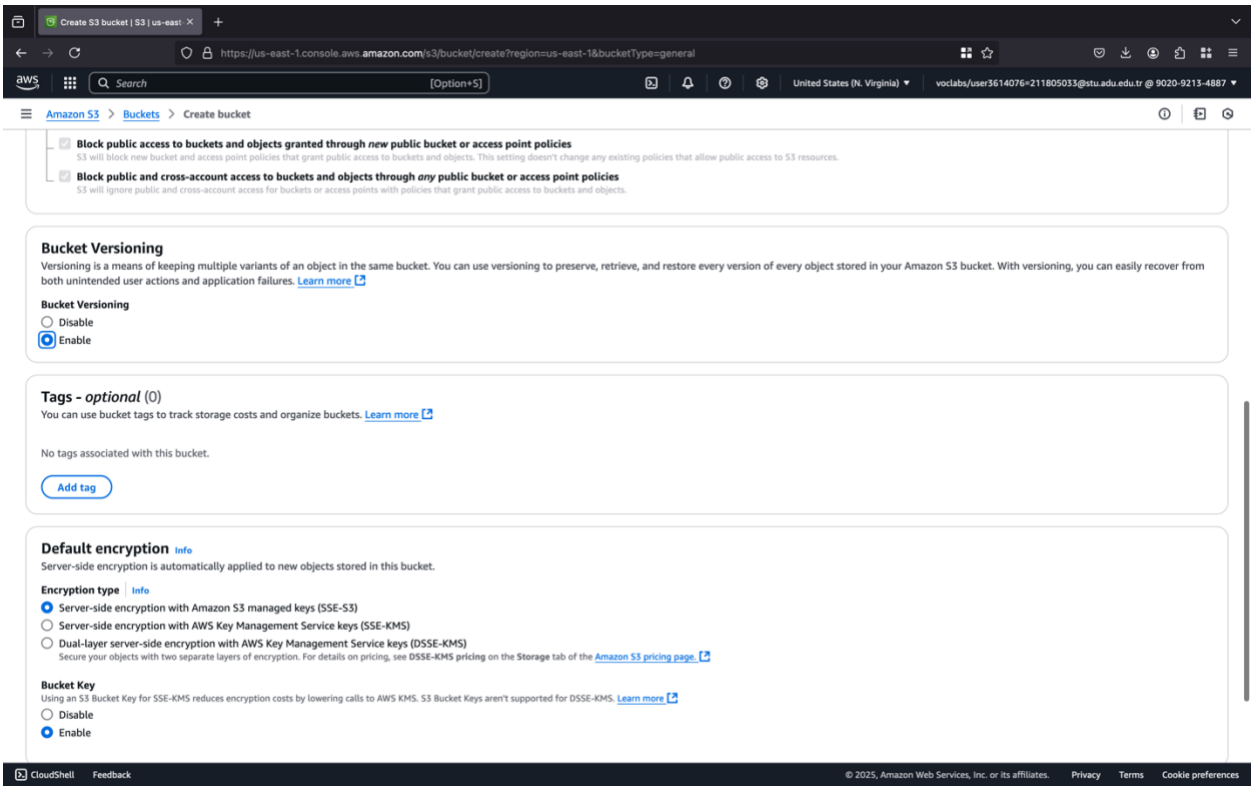


After multiple attempts to execute code in AWS SageMaker, I encountered limitations that prevented successful execution. This was primarily due to using the AWS Learner Lab environment, which imposes certain restrictions on resource availability and service usage. These constraints made it challenging to fully utilize SageMaker's capabilities for training and deploying machine learning models during my initial hands-on exploration.

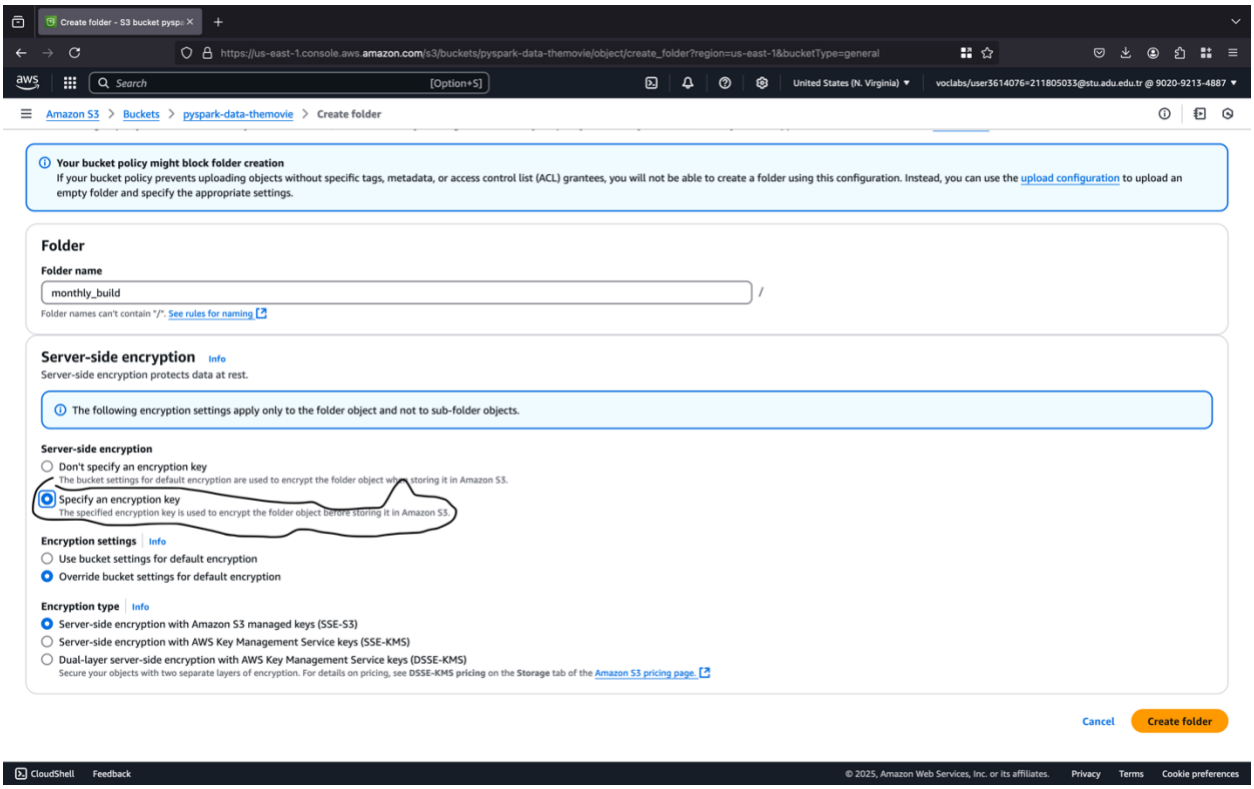


Then I tried to reconfigure the permisions to be able to use sagemaker but no matter what I did I couldn't use them.

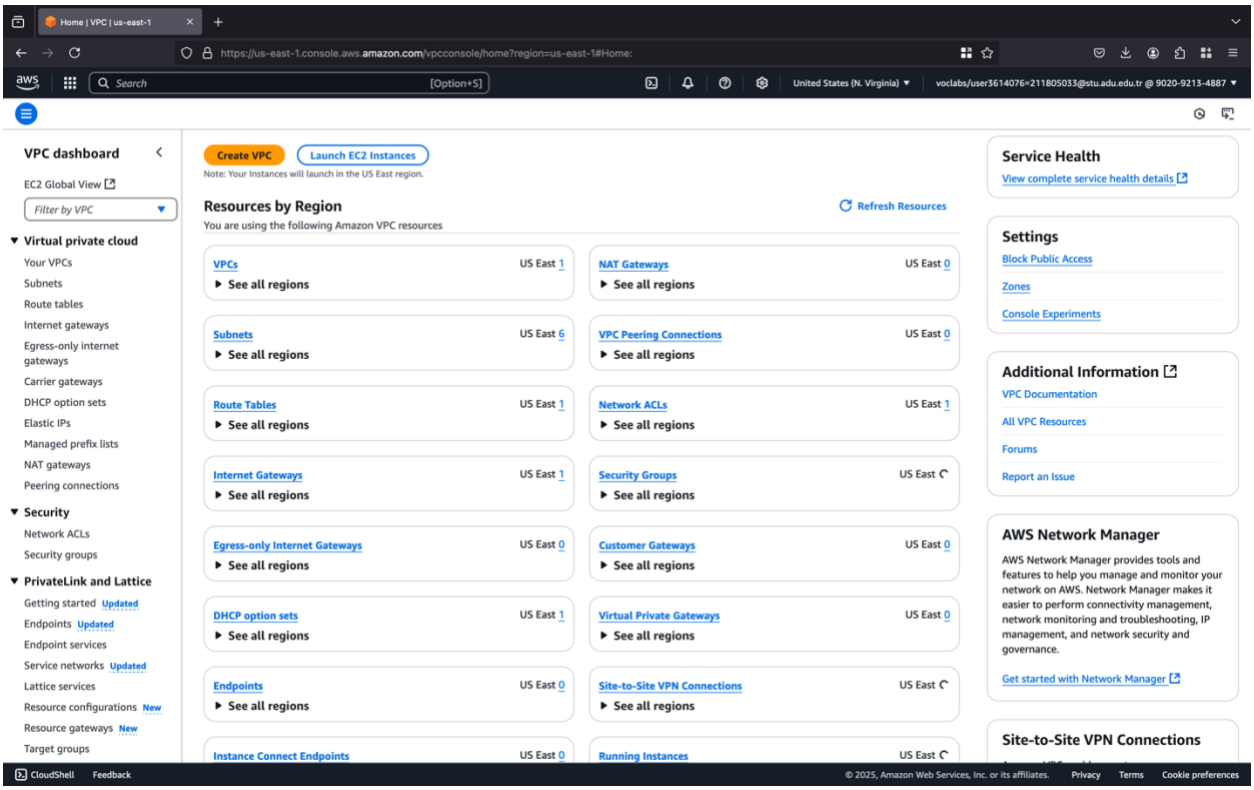
Determined to overcome the challenges I faced with the AWS Learner Lab, I invested significant effort into researching solutions. I read extensive documentation and watched numerous tutorials to better understand AWS SageMaker and other services. The idea of not being able to execute even a basic example after all this research motivated me further. To circumvent the limitations of the Learner Lab, I decided to switch to AWS Free Tier, allowing me to experiment with a broader range of services and configurations without restrictions.



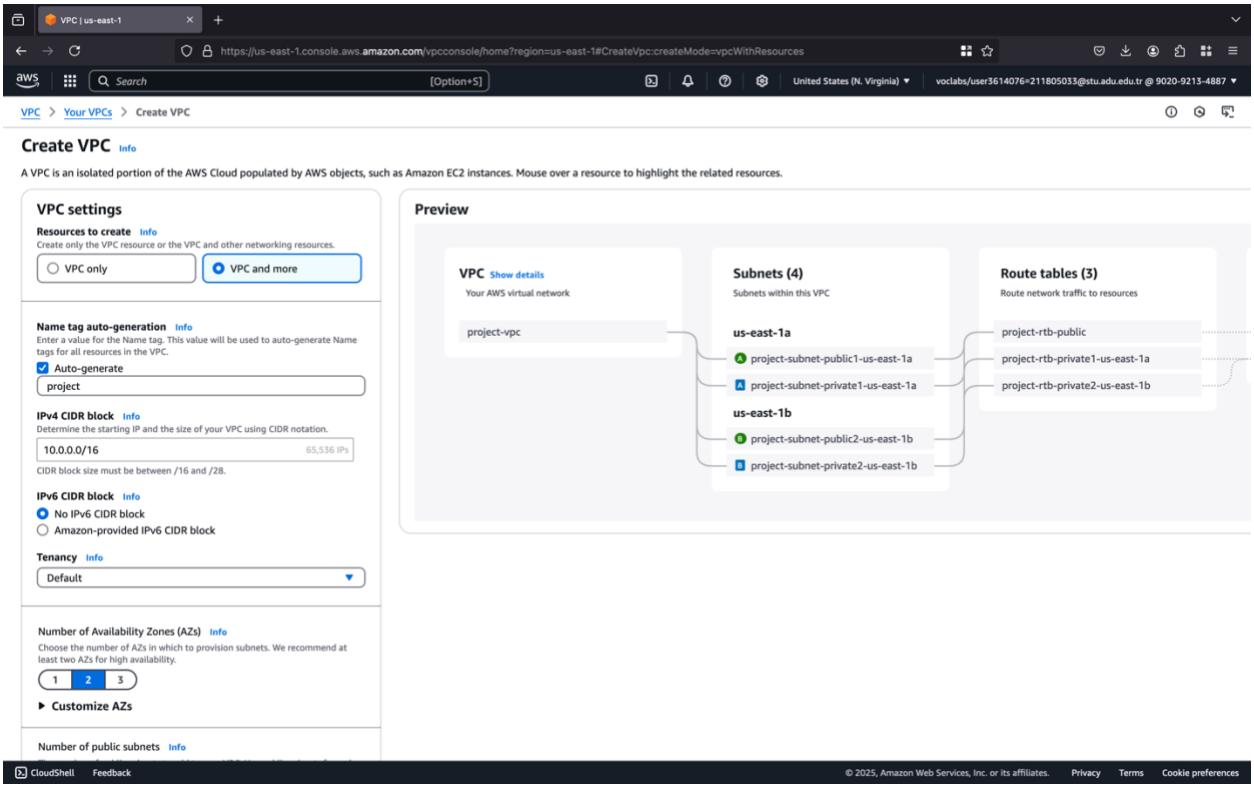
I create a s3 bucket again for data.



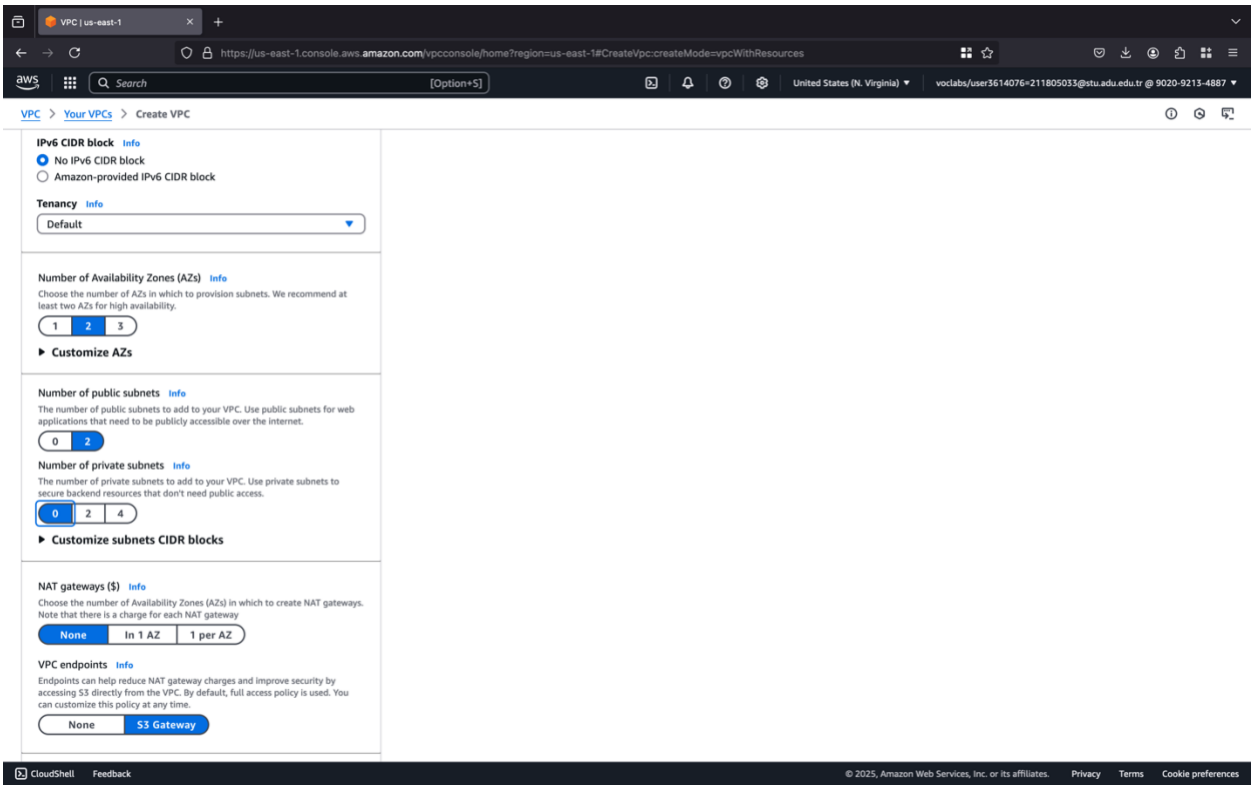
I specified an encryption key to encrypt the folder in Amazon s3



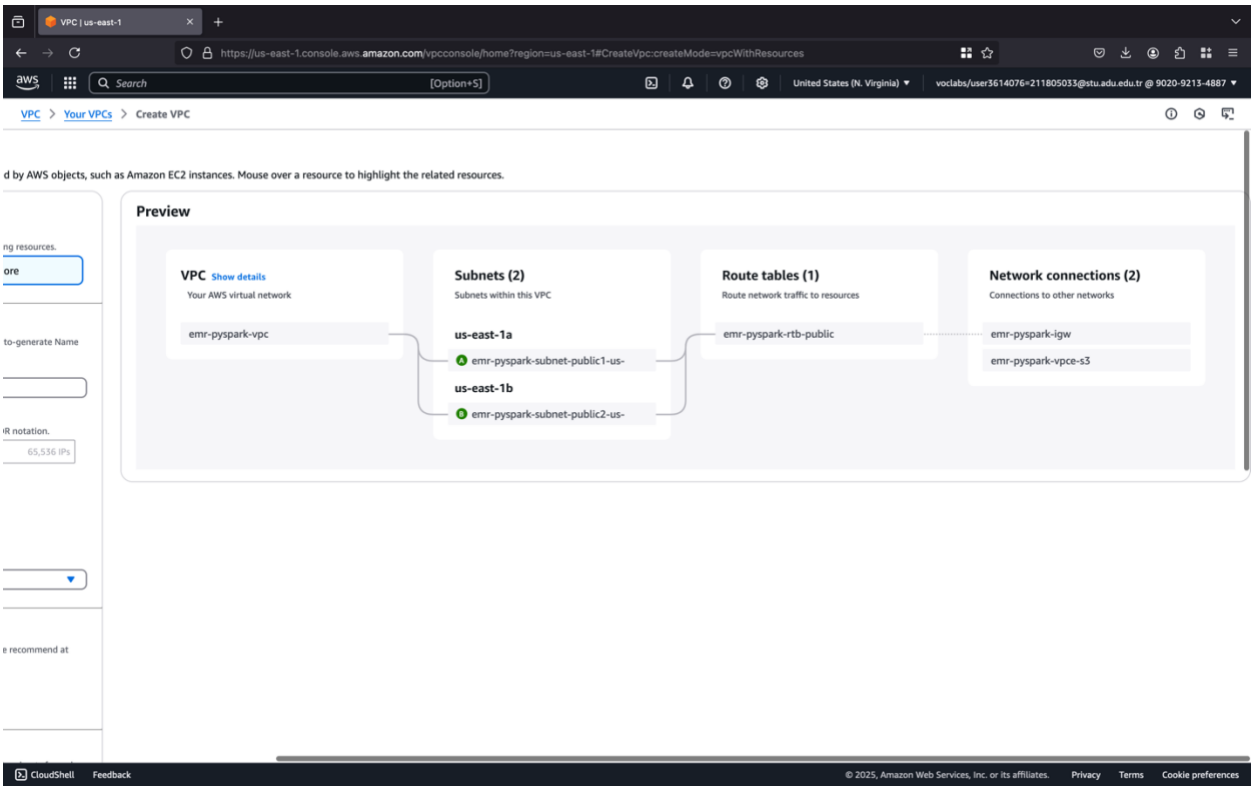
To ensure a secure and reliable networking environment for my AWS infrastructure, I utilized Amazon Virtual Private Cloud (VPC). VPC allows me to create an isolated network within the AWS cloud, giving me full control over network configuration, including subnets, routing, and security settings. By using VPC, I could implement enhanced security measures, such as custom security groups and network access control lists (ACLs), while ensuring that resources within the VPC could communicate efficiently and securely.



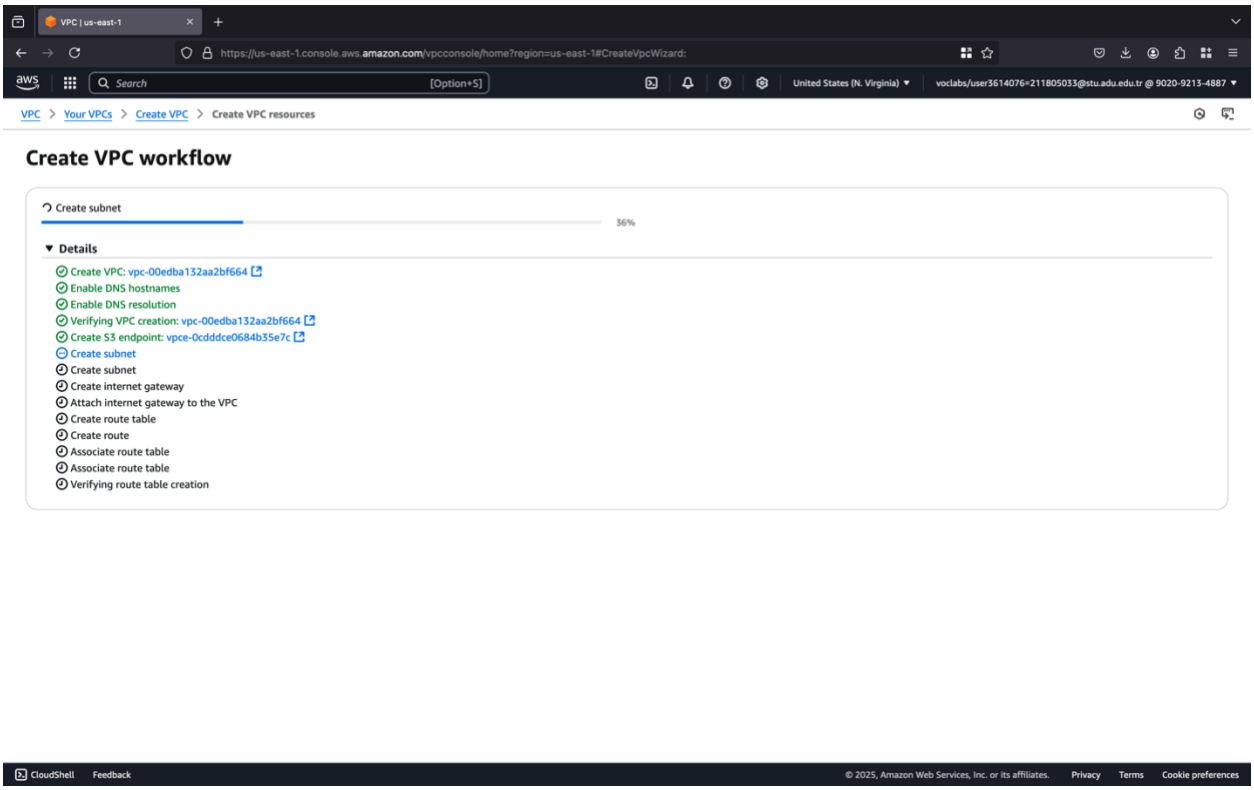
Here I show What configurations I did for vpc.



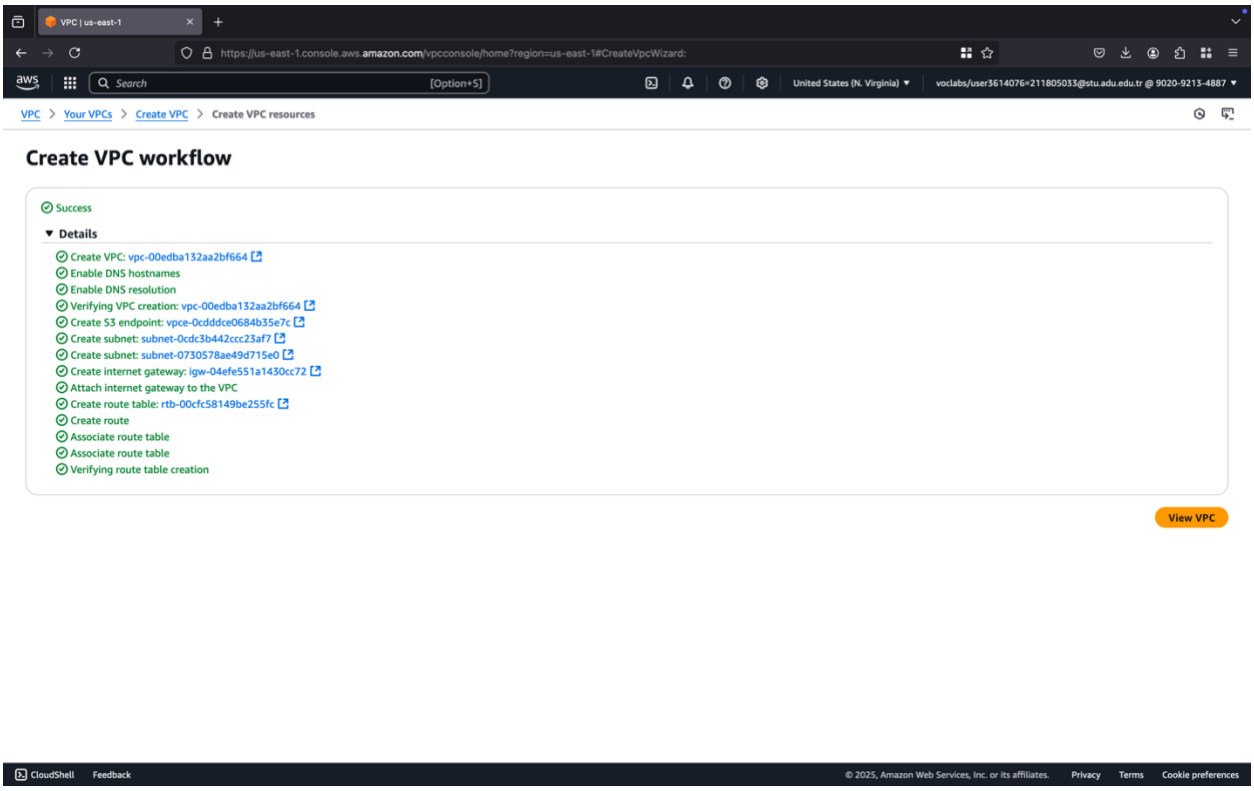
I choose 2 numbers of zones as default 2 number of public subnets.



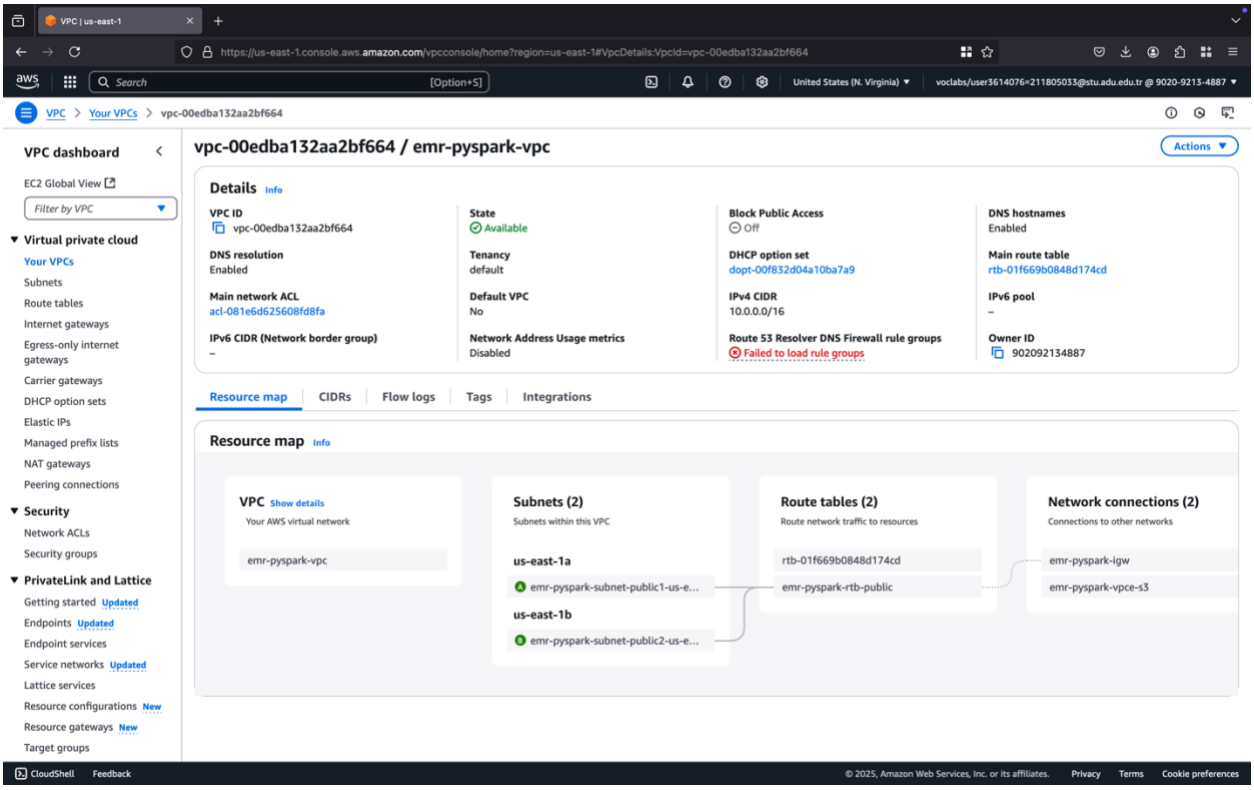
And this is fianl schema of VPC .



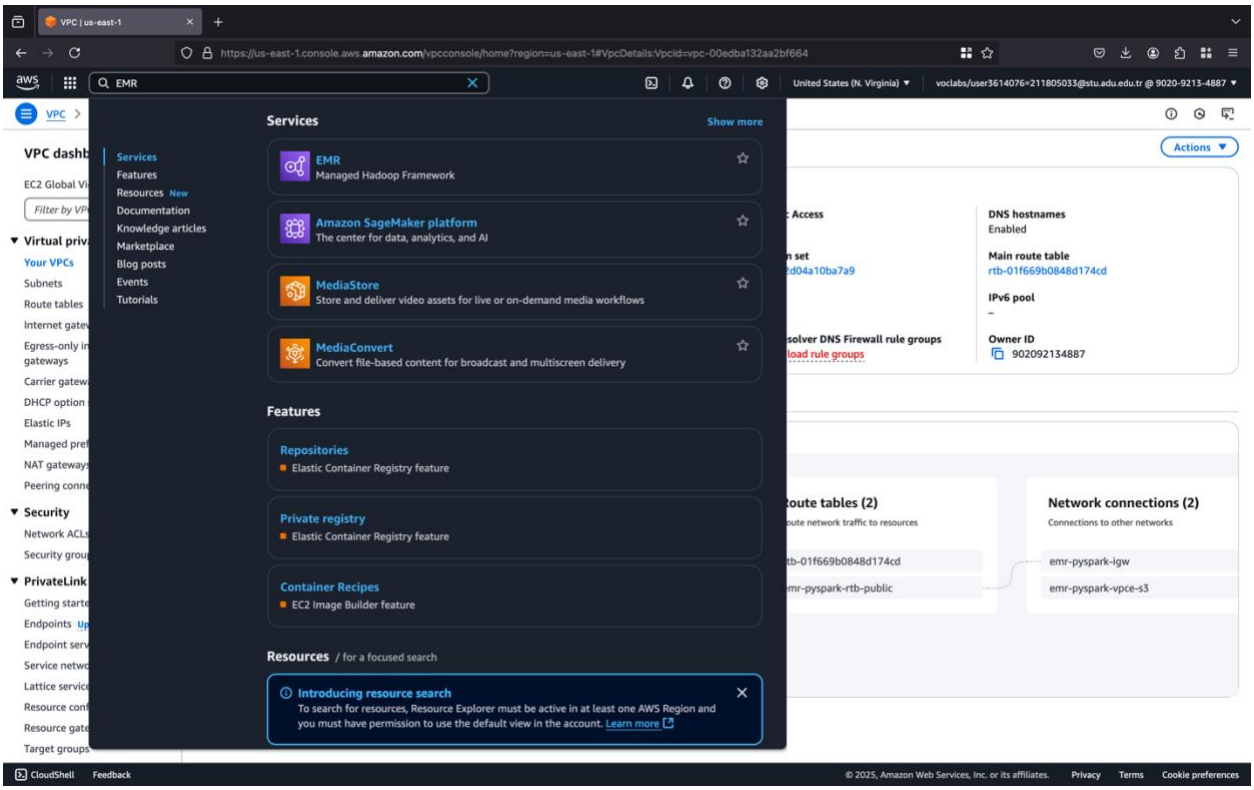
I hit the create button.

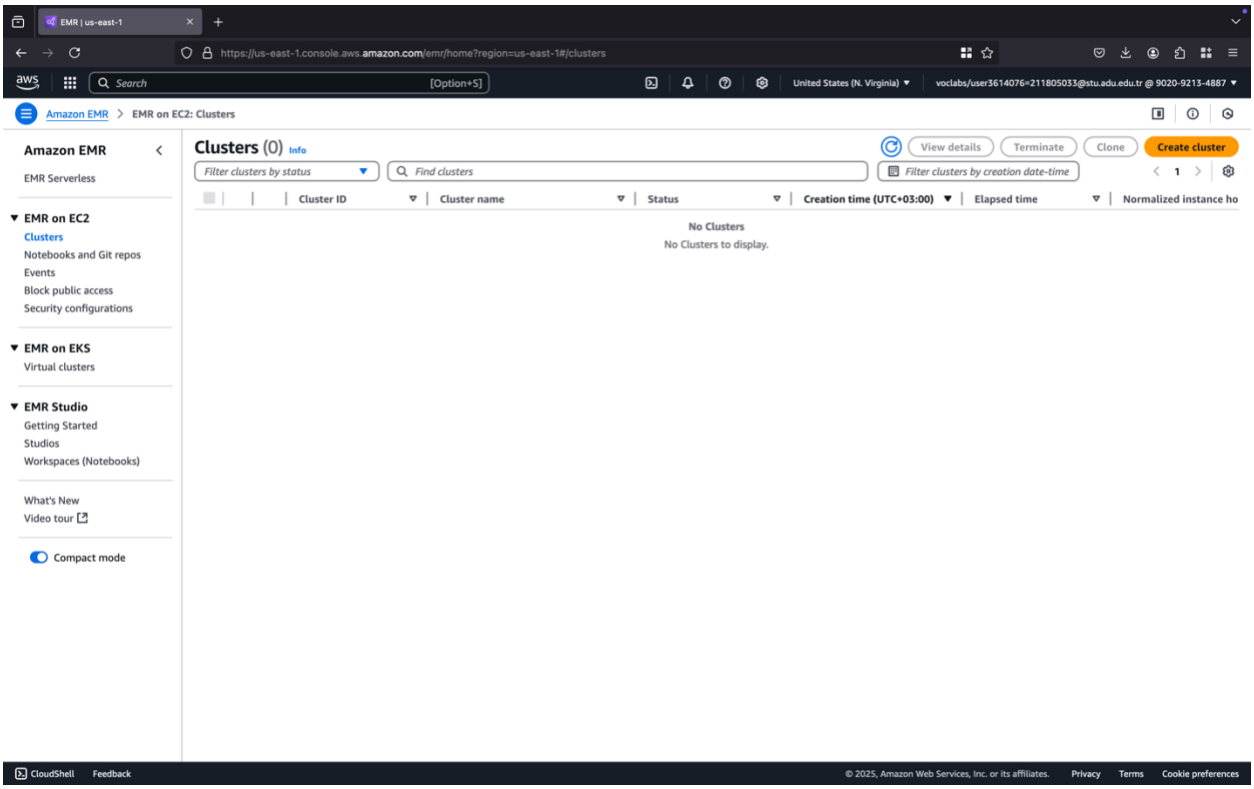


The VPC workflow created.

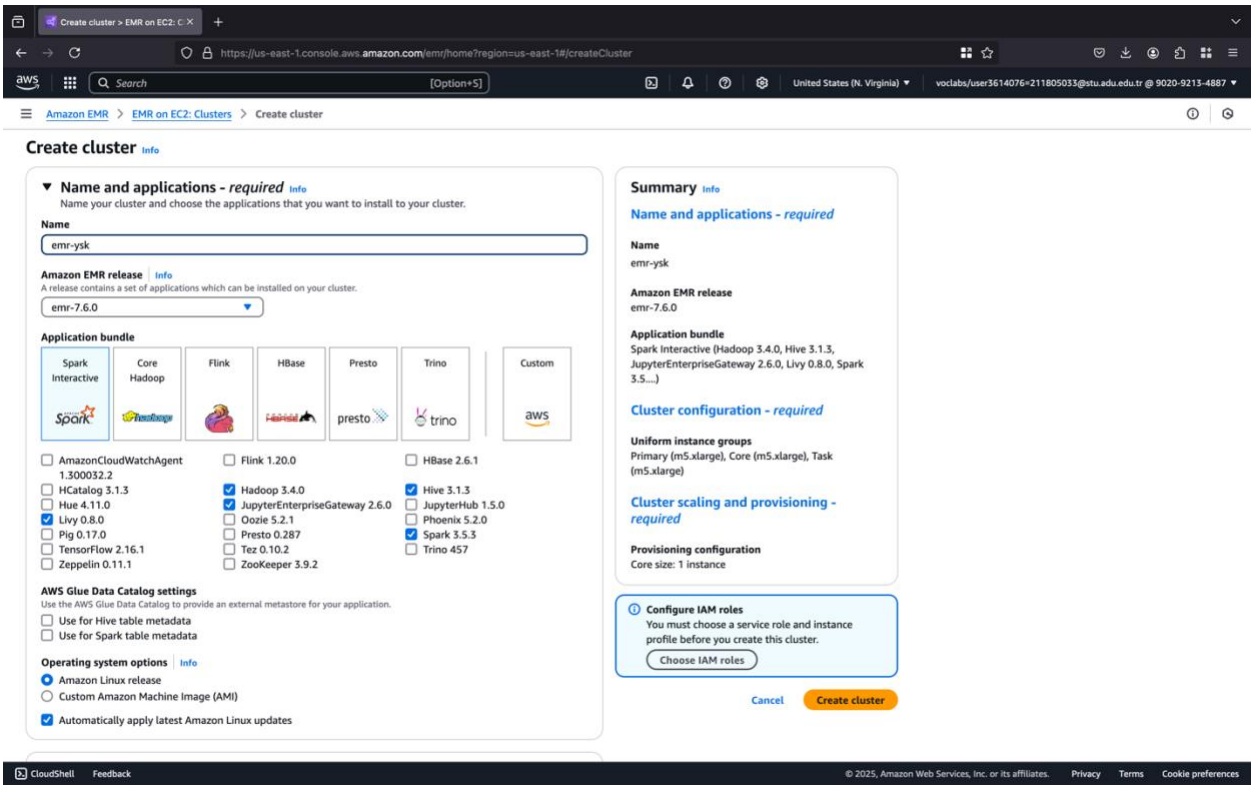


As part of my continued exploration of AWS services, I conducted in-depth research on Amazon EMR (Elastic MapReduce), a managed cluster platform designed for processing large amounts of data using frameworks like Hadoop and Spark. Recognizing its potential for scalable big data processing, I began hands-on work with EMR, configuring clusters to run distributed data processing tasks. This allowed me to further enhance my machine learning pipeline and gain experience with distributed computing in the AWS ecosystem.

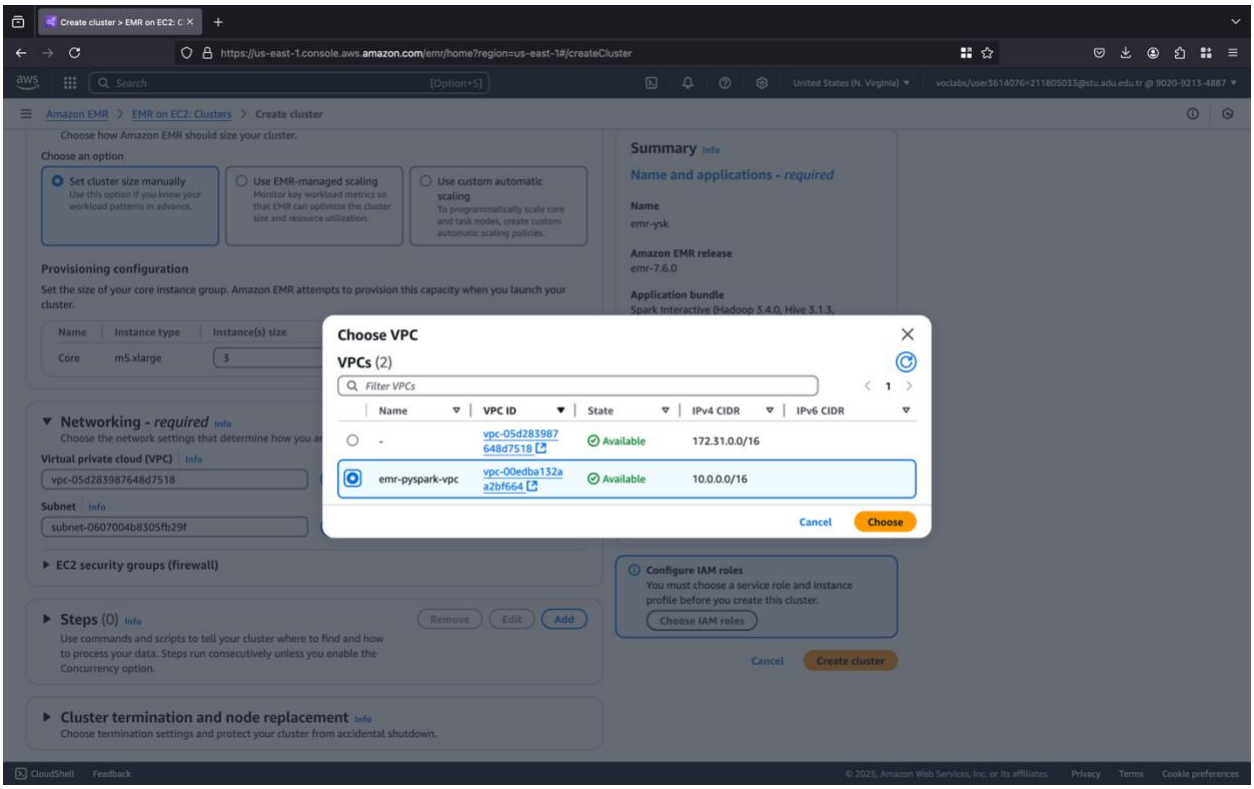




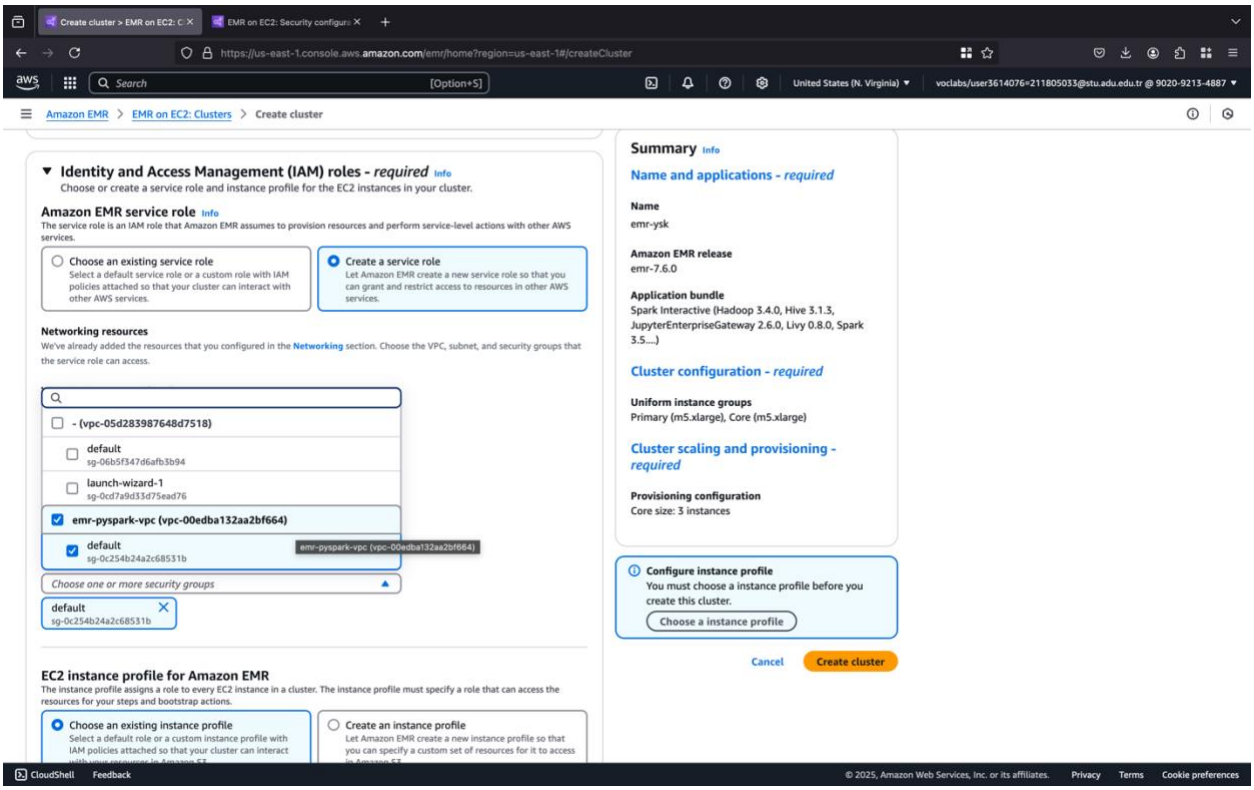
I hit the create cluster button to create the emr cluster.

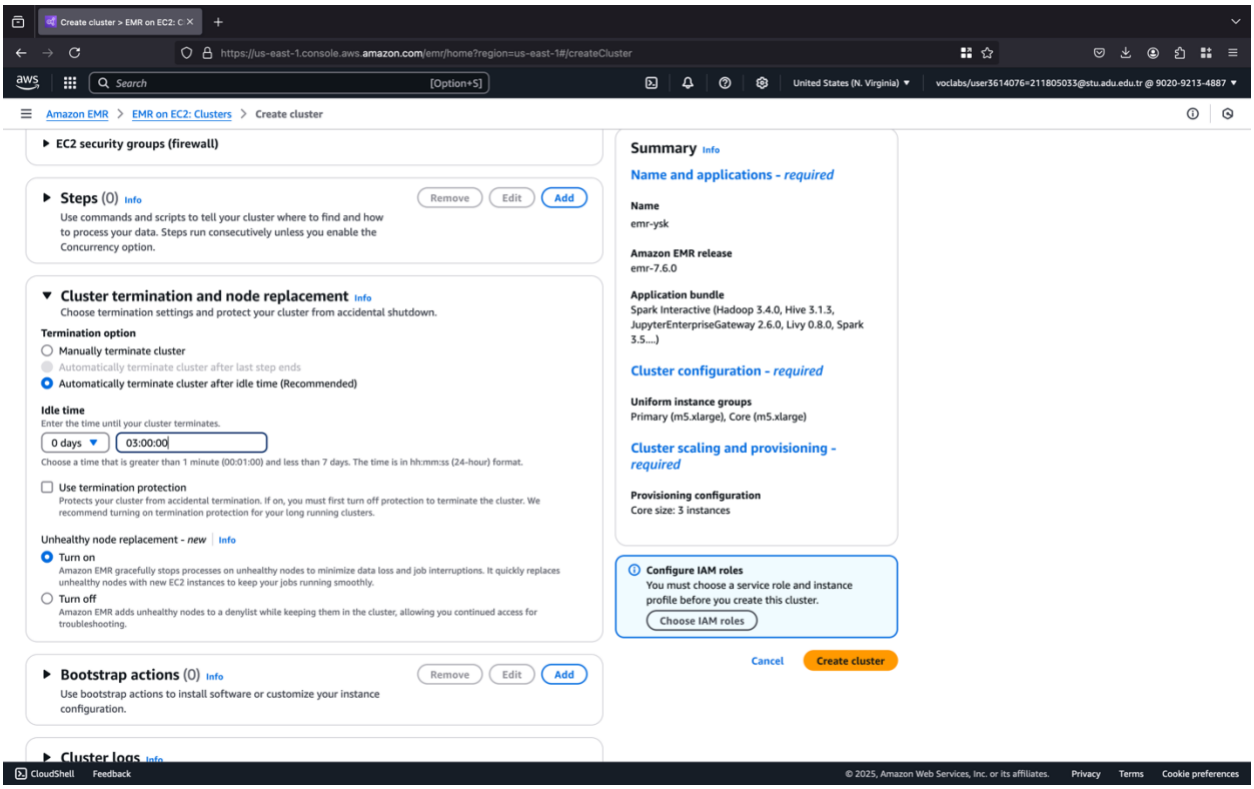


Configuration for emr of cource I Choose pyspark.

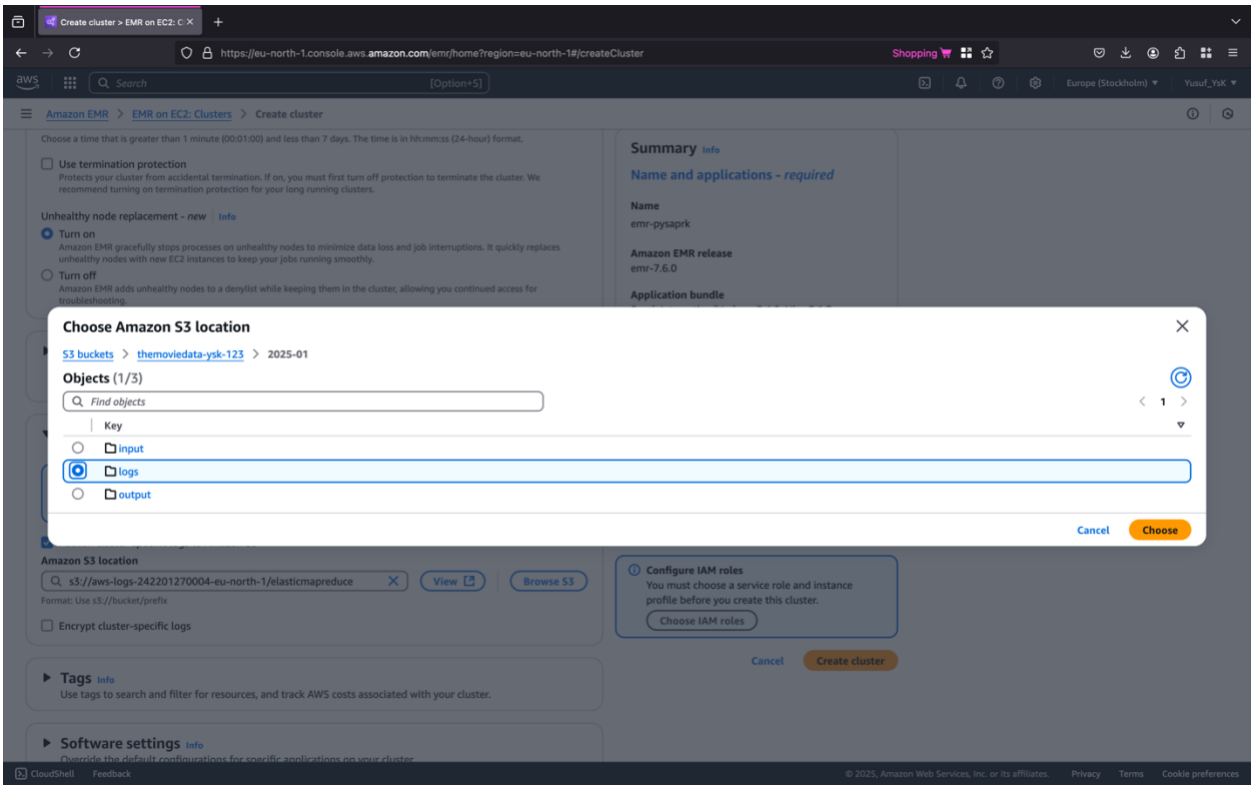


I choose the vpc I created for. Networking.

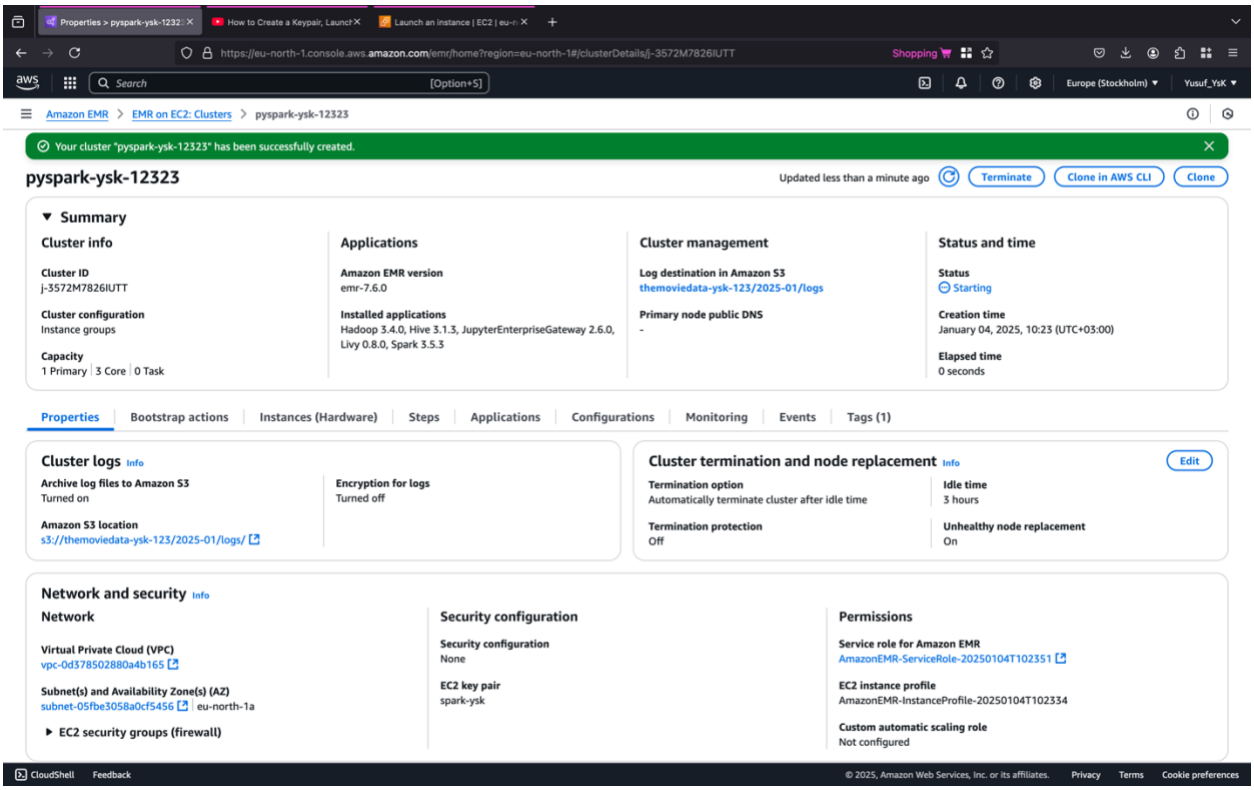




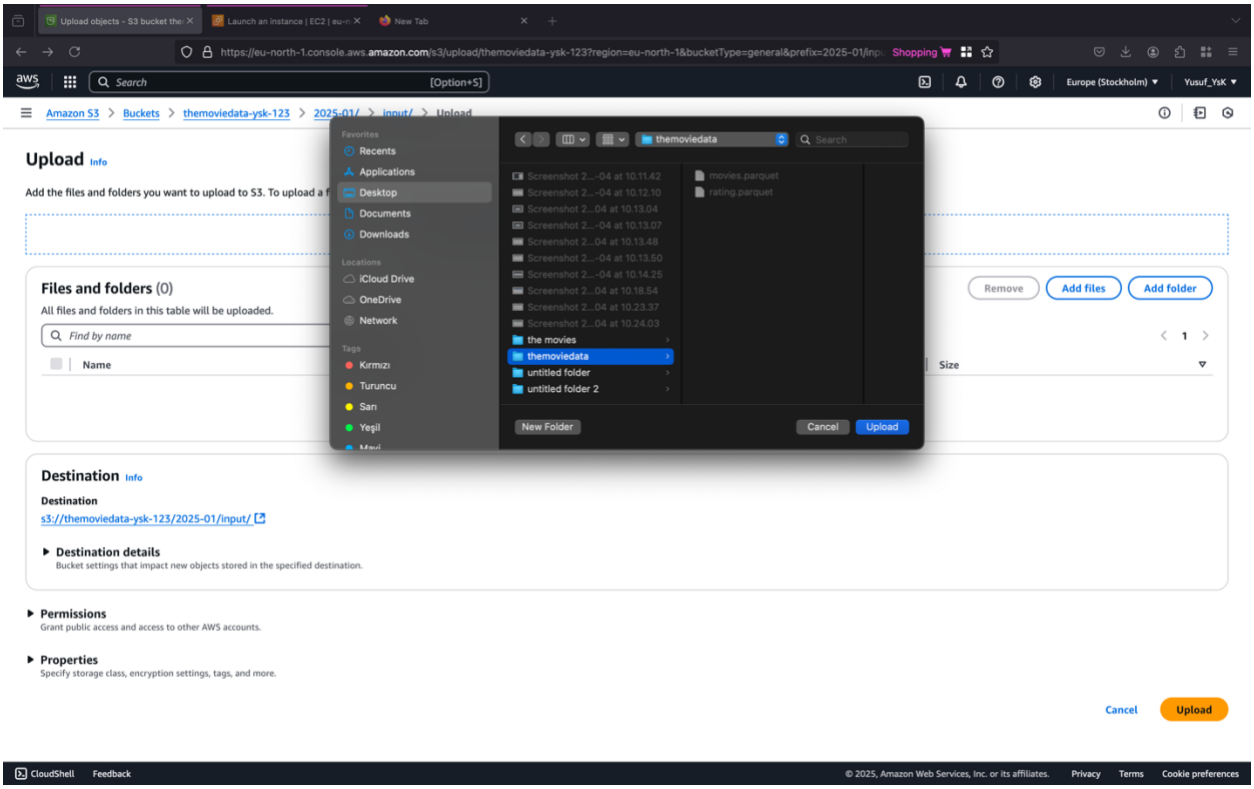
To manage costs effectively and avoid unexpected charges, I set the termination time for my Amazon EMR clusters to 3 hours. This decision was made after registering my credit card with AWS, as I wanted to ensure I wouldn't incur excessive costs. Since AWS operates on a pay-per-use model, where you are billed based on the resources you consume, I made it a priority to control the duration for which the resources are active, minimizing any potential overages.



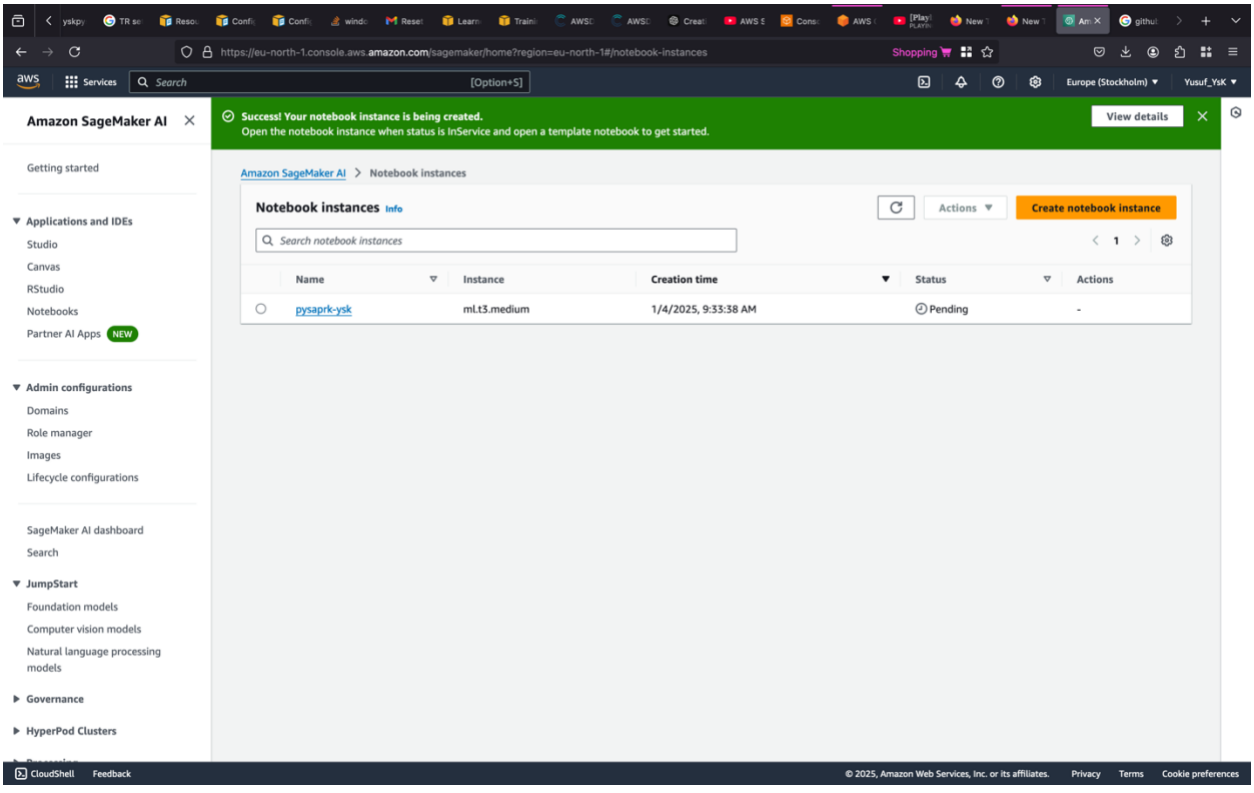
I choose s3 bucket location to show logs in emr.



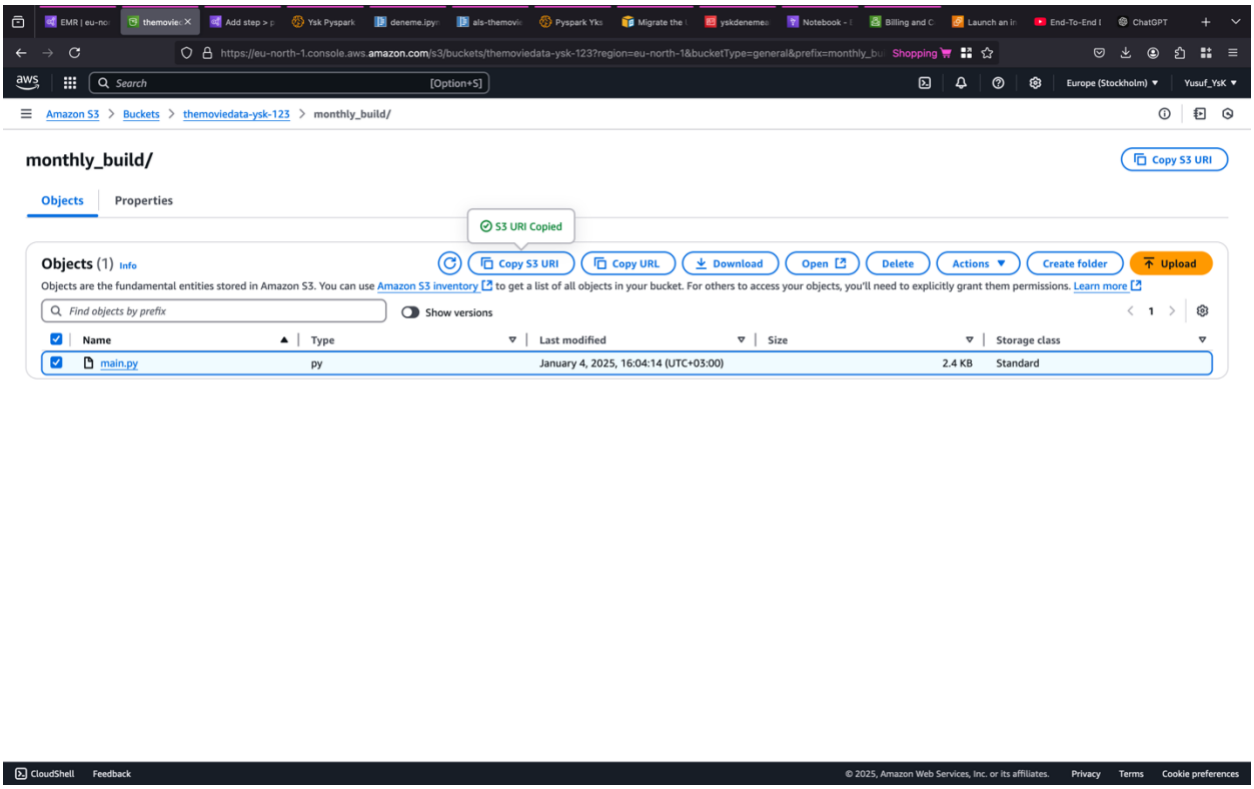
After configuring the necessary settings and resources, I successfully created my EMR cluster. This marked a key milestone in my project, as it enabled me to leverage the power of distributed computing for processing large datasets. With the cluster up and running, I could begin executing data processing tasks using frameworks like Hadoop or Spark, facilitating the next steps in my machine learning pipeline.



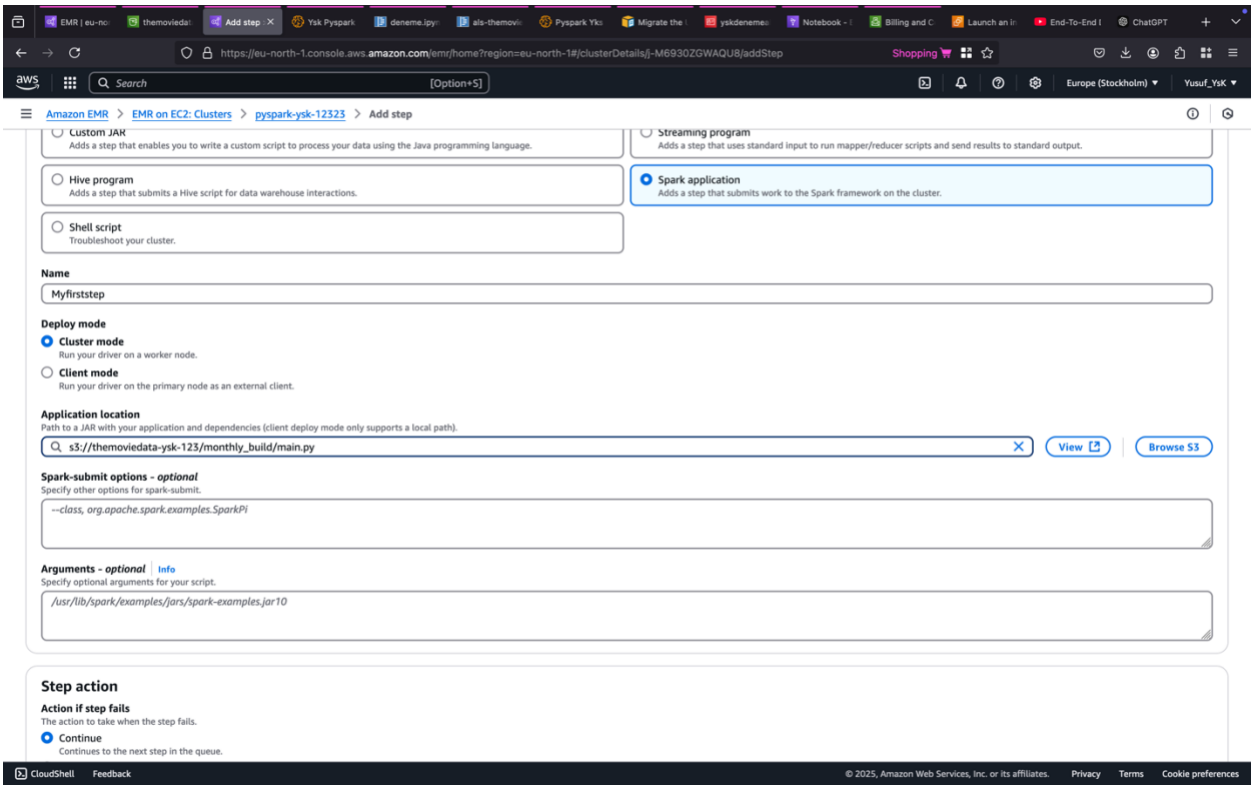
I load the moviedata which I will use for emr to s3 bucket.



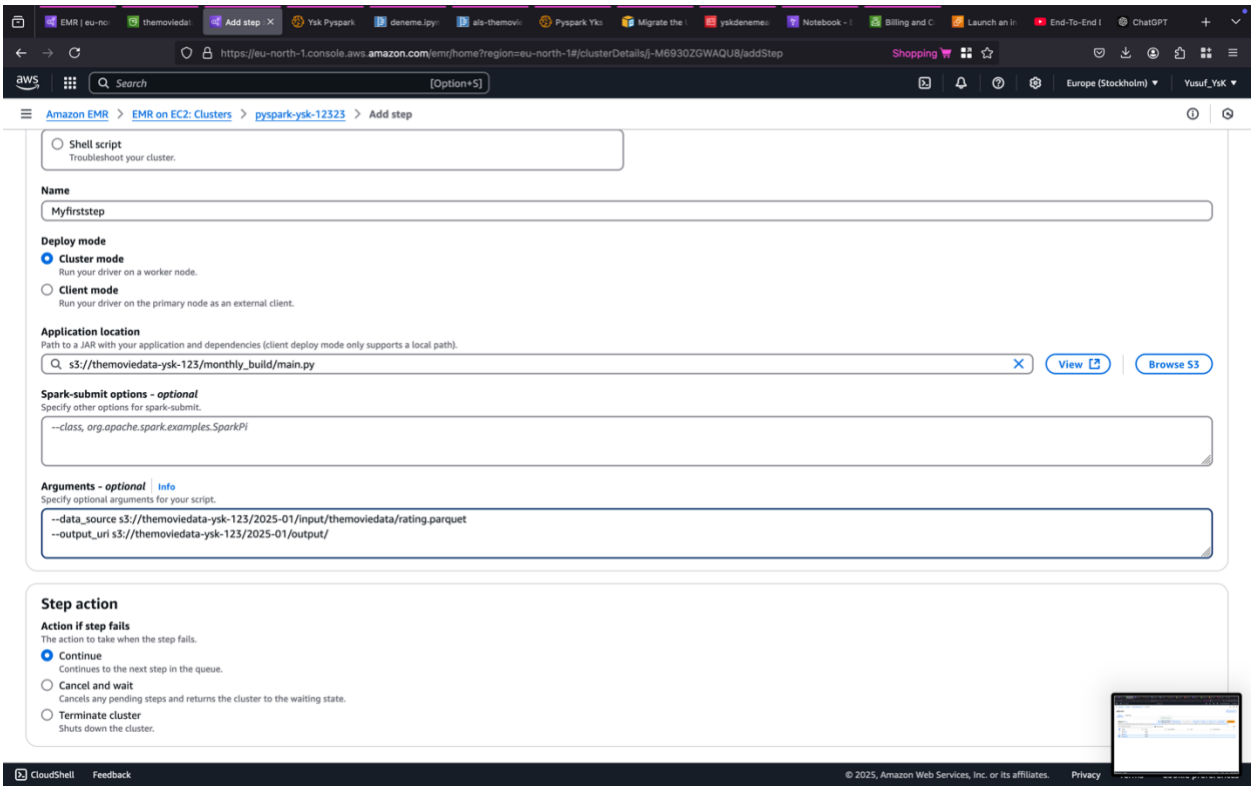
I wait to emr launch.

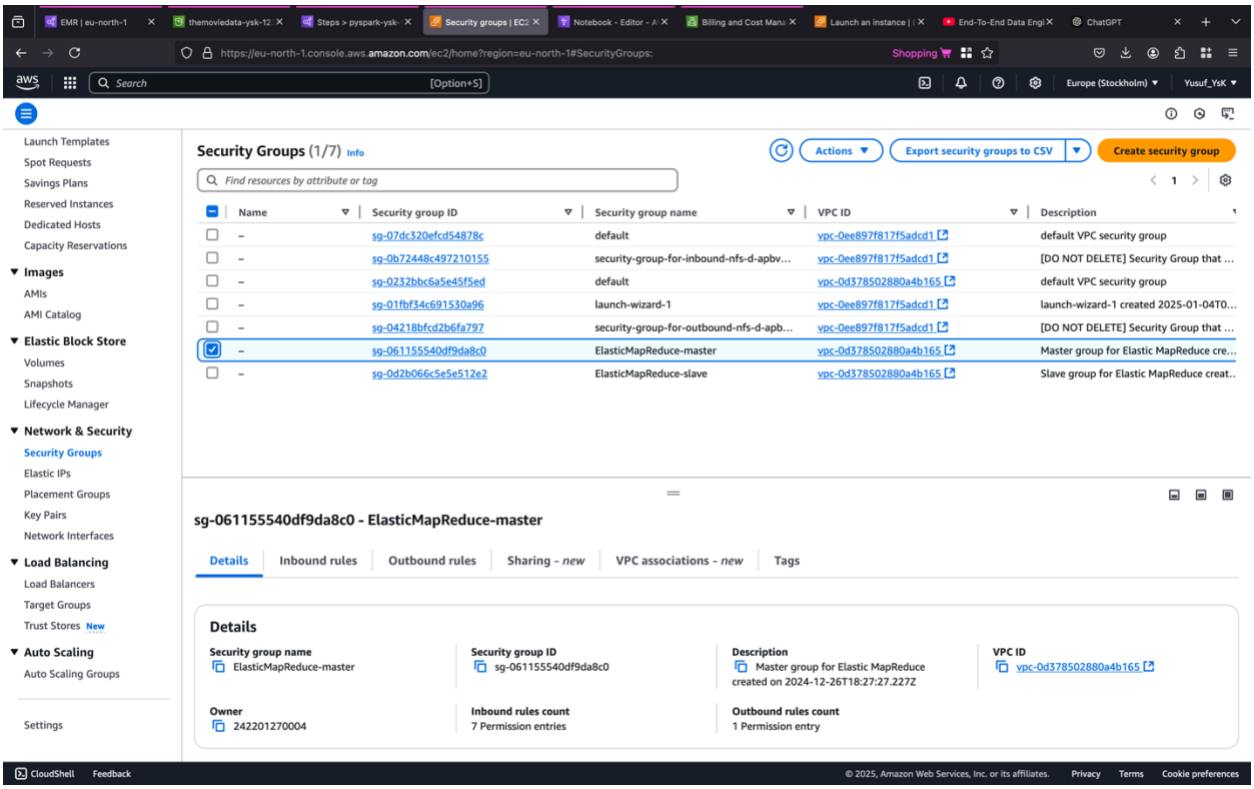


Within the monthly_build folder, I will store the Python code responsible for interacting with the data stored in the S3 bucket. This code will handle tasks such as loading the dataset into the EMR cluster, preprocessing the data, and training the machine learning model. By organizing the code in this folder, I can keep the workflow streamlined and ensure that the necessary scripts are readily available for execution as part of my data processing pipeline.

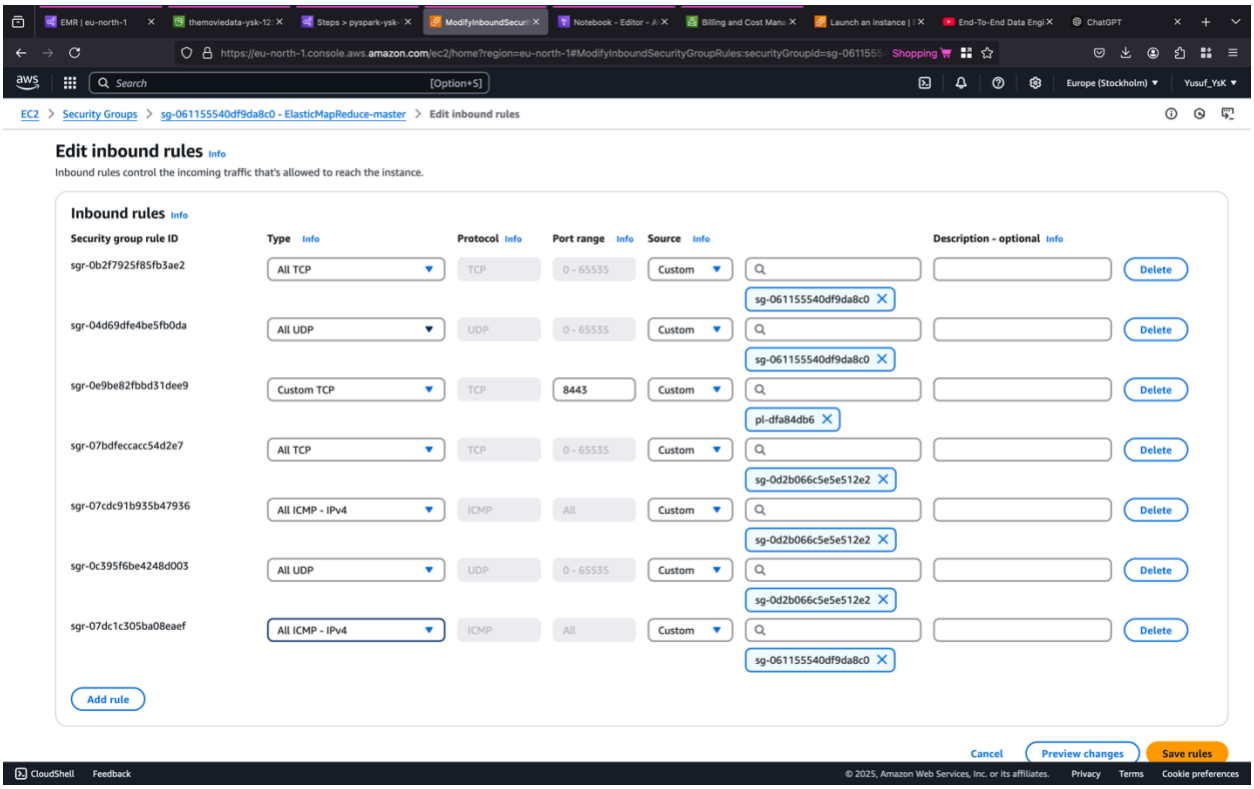


My first step of emr

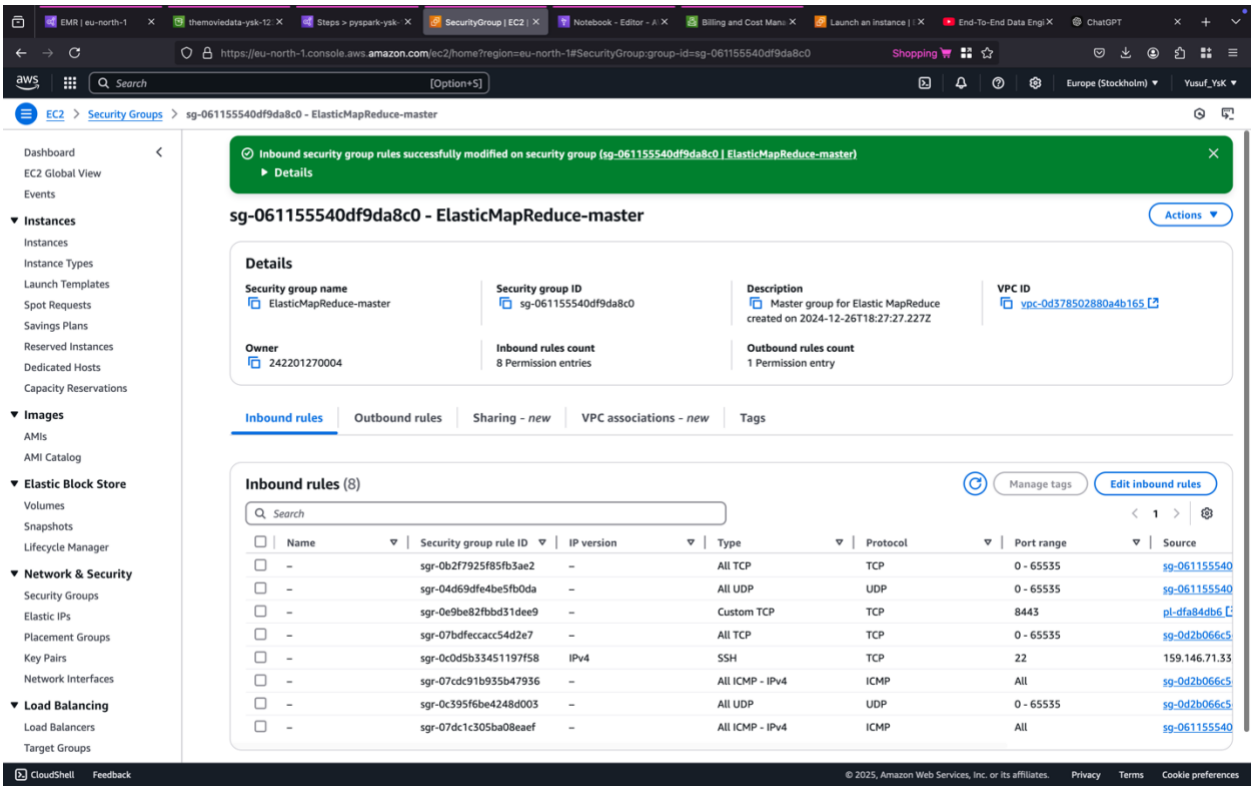




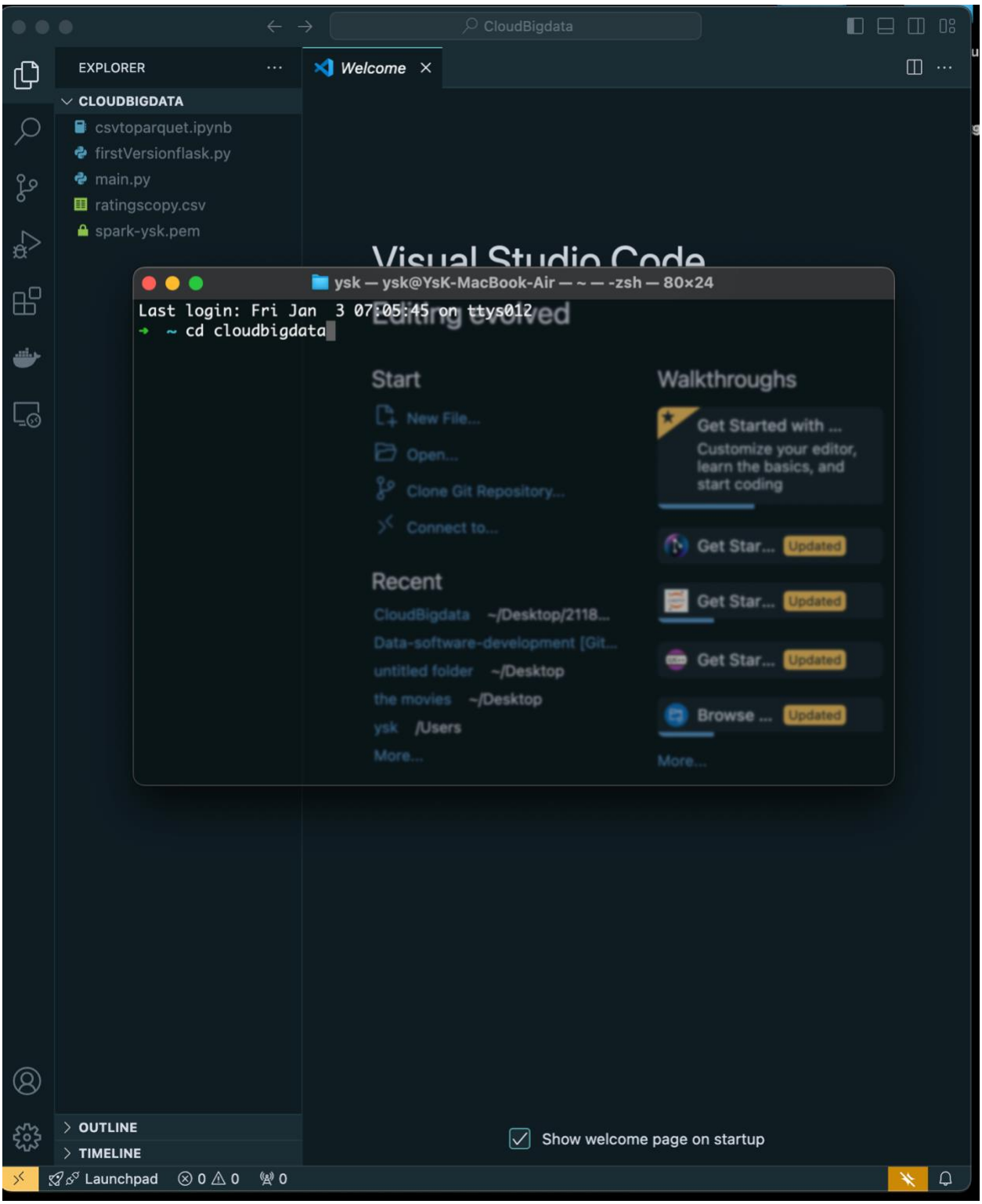
I click on the Security group id to reconfigure network option to be able to connect emr via ssh.



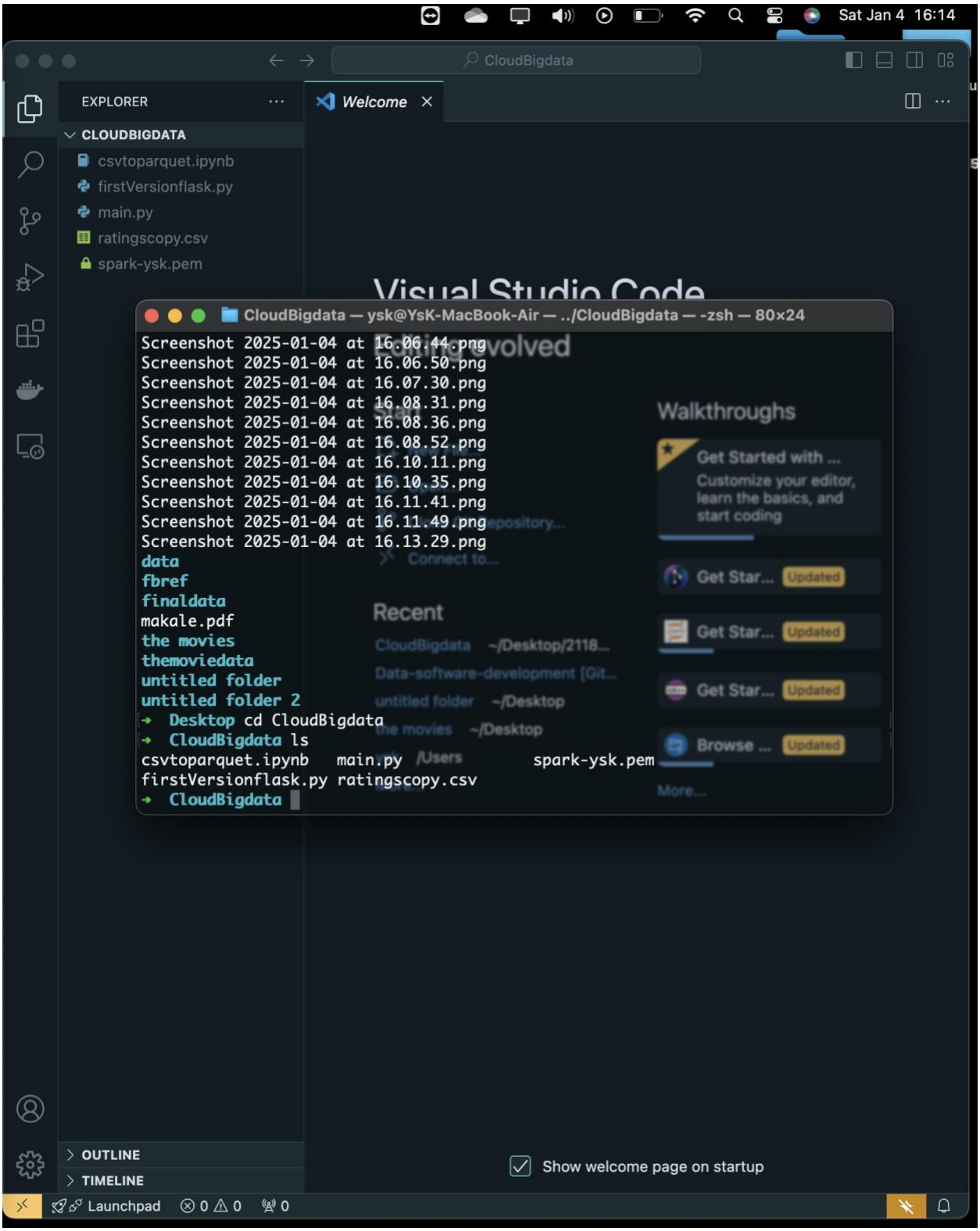
Here I add rule choose type to be ssh and source to be my ip



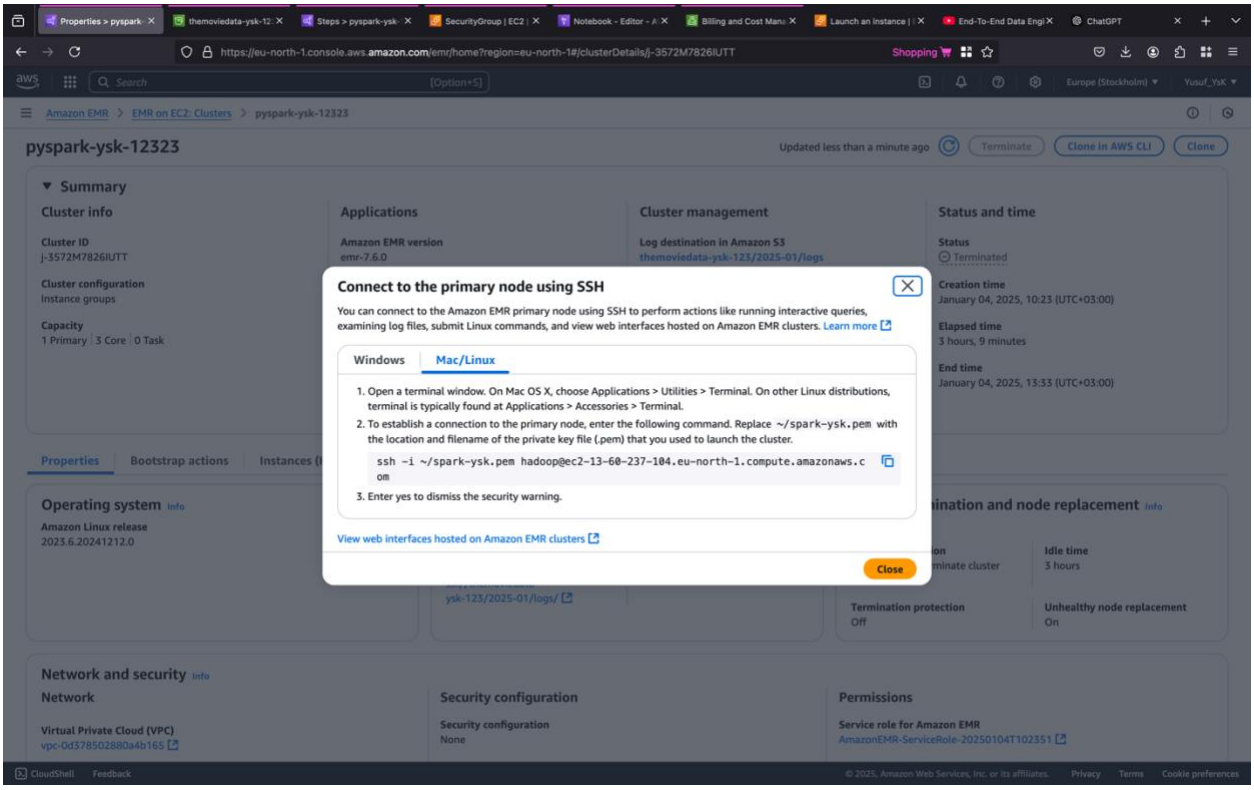
I sucsselfully reconfigure elastic map reduce master node.

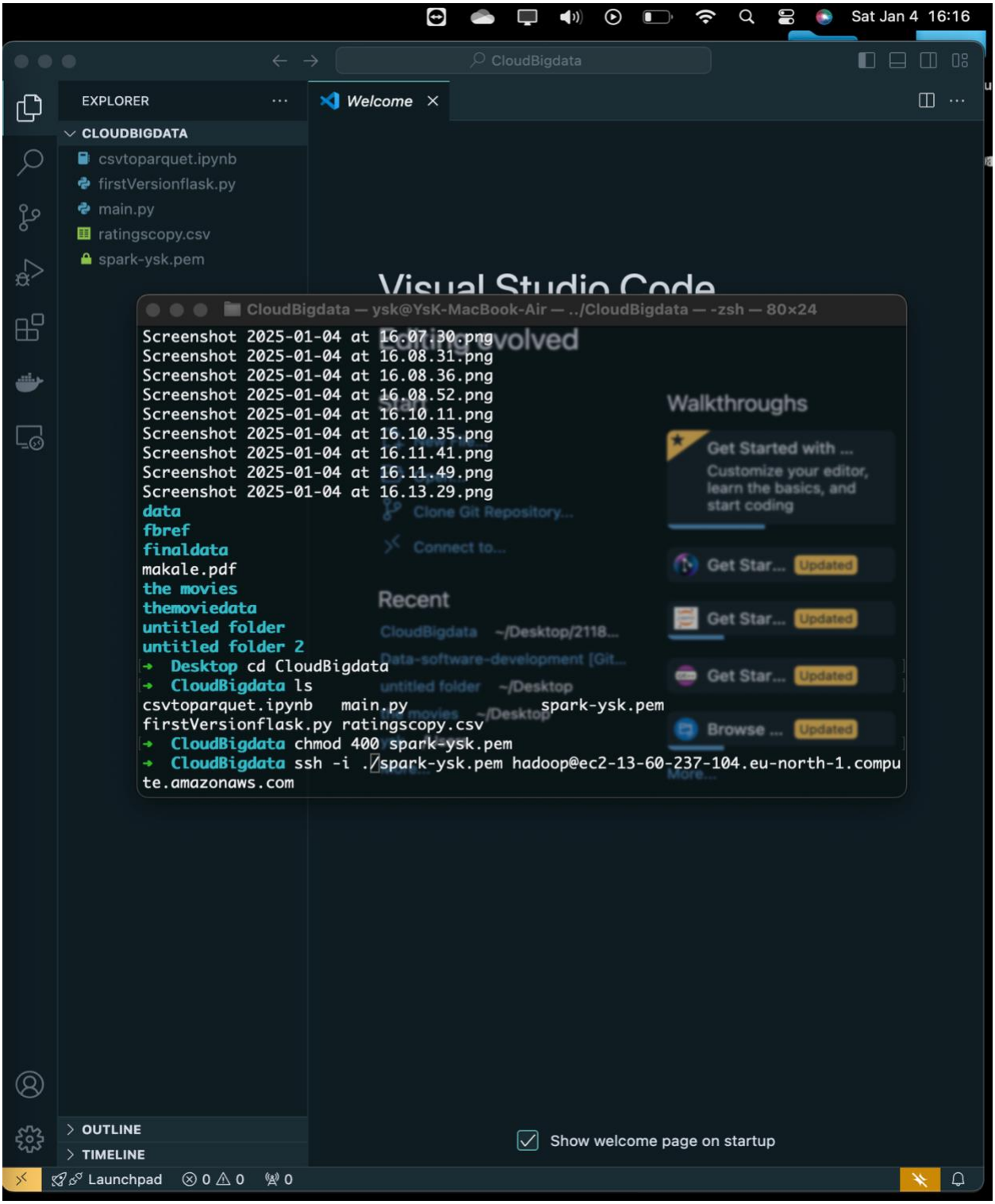


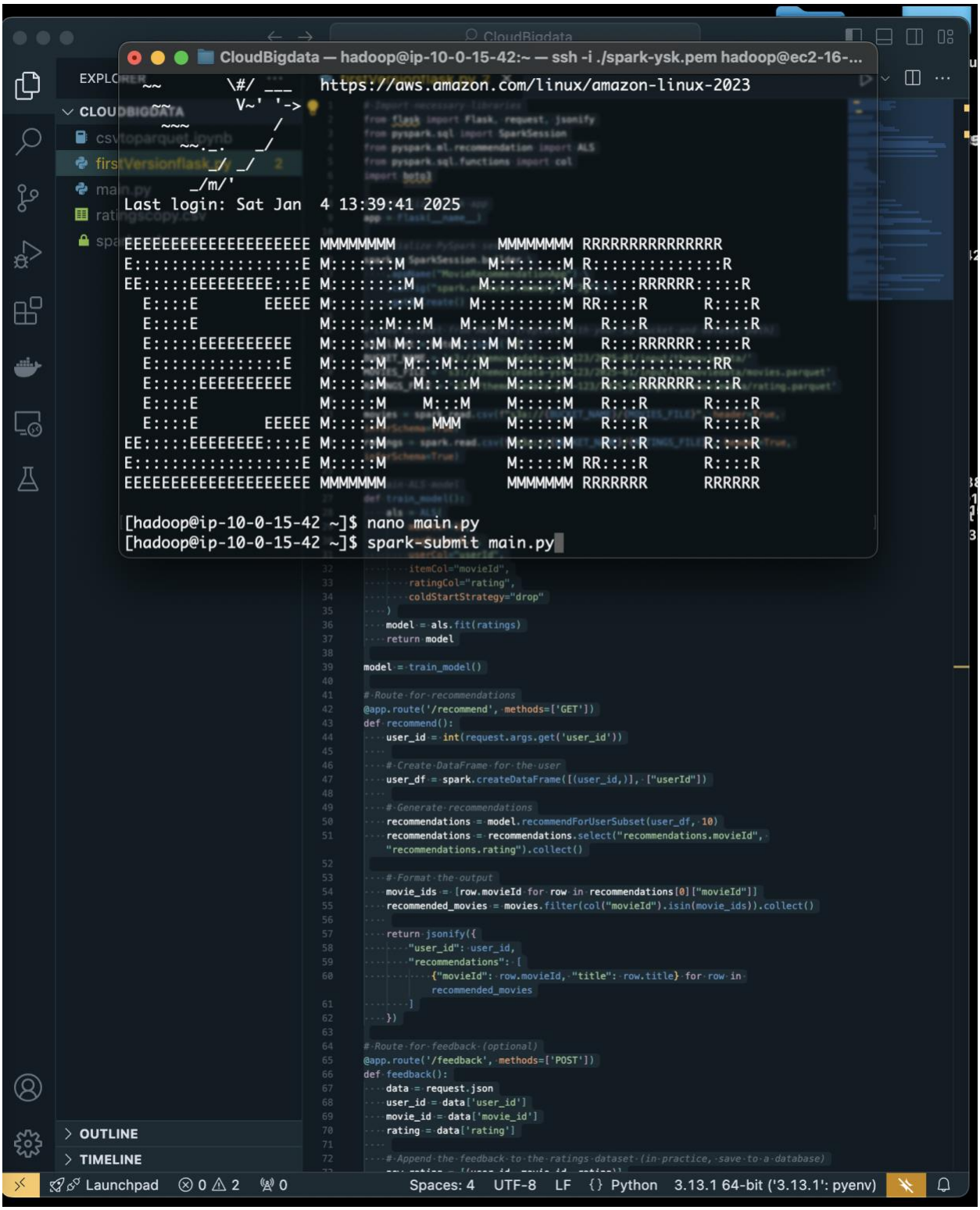
My loacal terminal to use ssh


Sat Jan 4 16:14
CloudBigdata
EXPLORER
CLOUDBIGDATA
csvtoparquet.ipynb
firstVersionflask.py
main.py
ratingscopy.csv
spark-ysk.pem
Welcome
Visual Studio Code
Editing evolved
Walkthroughs
Get Started with ...
Customize your editor, learn the basics, and start coding
Get Star... Updated
Get Star... Updated
Get Star... Updated
Browse ... Updated
More...
Recent
CloudBigdata ~/Desktop/2118...
Data-software-development [Git...
untitled folder ~/Desktop
the movies ~/Desktop
Show welcome page on startup
OUTLINE
TIMELINE
Launchpad
CloudBigdata — ysk@YsK-MacBook-Air — ../CloudBigdata — -zsh — 80×24
Screenshot 2025-01-04 at 16.06.44.png
Screenshot 2025-01-04 at 16.06.50.png
Screenshot 2025-01-04 at 16.07.30.png
Screenshot 2025-01-04 at 16.08.31.png
Screenshot 2025-01-04 at 16.08.36.png
Screenshot 2025-01-04 at 16.08.52.png
Screenshot 2025-01-04 at 16.10.11.png
Screenshot 2025-01-04 at 16.10.35.png
Screenshot 2025-01-04 at 16.11.41.png
Screenshot 2025-01-04 at 16.11.49.png
Screenshot 2025-01-04 at 16.13.29.png
data
fbref
finaldata
makale.pdf
the movies
themoviedata
untitled folder
untitled folder 2
→ Desktop cd CloudBigdata
→ CloudBigdata ls
csvtoparquet.ipynb main.py /Users spark-ysk.pem
firstVersionflask.py ratingscopy.csv
→ CloudBigdata

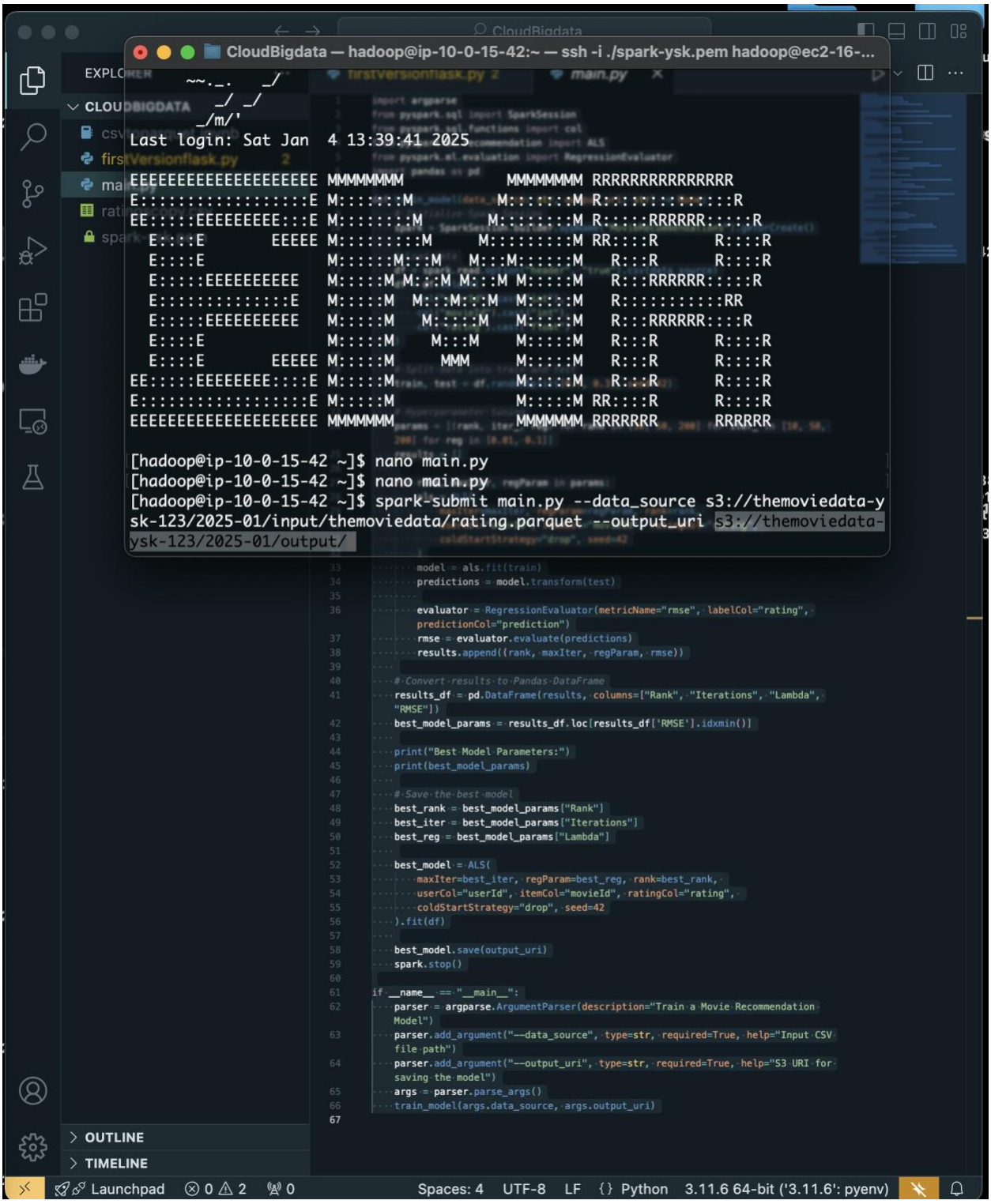
To securely access the EMR cluster, I used the PEM key pair that I generated when setting up my EC2 instances. This key pair allows for SSH access, ensuring that the connection to the cluster is encrypted and secure. By leveraging SSH with the PEM key, I was able to remotely access the EMR cluster, execute commands, and manage the data processing tasks directly from my local machine.




Sat Jan 4 16:16
CloudBigdata
EXPLORER
Welcome
CLOUDBIGDATA
csvtoparquet.ipynb
firstVersionflask.py
main.py
ratingscopy.csv
spark-ysk.pem
Visual Studio Code
Walkthroughs
Get Started with ...
Customize your editor, learn the basics, and start coding
Get Star... Updated
Get Star... Updated
Get Star... Updated
Browse ... Updated
Recent
CloudBigdata ~/Desktop/2118...
Data-software-development [Git...
untitled folder ~/Desktop
Clone Git Repository...
Connect to...
Show welcome page on startup
OUTLINE
TIMELINE
Launchpad
CloudBigdata — ysk@YsK-MacBook-Air — ../CloudBigdata — -zsh — 80×24
Screenshot 2025-01-04 at 16.07.30.png
Screenshot 2025-01-04 at 16.08.31.png
Screenshot 2025-01-04 at 16.08.36.png
Screenshot 2025-01-04 at 16.08.52.png
Screenshot 2025-01-04 at 16.10.11.png
Screenshot 2025-01-04 at 16.10.35.png
Screenshot 2025-01-04 at 16.11.41.png
Screenshot 2025-01-04 at 16.11.49.png
Screenshot 2025-01-04 at 16.13.29.png
data
fbref
finaldata
makale.pdf
the movies
themoviedata
untitled folder
untitled folder 2
→ Desktop cd CloudBigdata
→ CloudBigdata ls
csvtoparquet.ipynb main.py spark-ysk.pem
firstVersionflask.py ratingscopy.csv
→ CloudBigdata chmod 400 spark-ysk.pem
→ CloudBigdata ssh -i ./spark-ysk.pem hadoop@ec2-13-60-237-104.eu-north-1.compute.amazonaws.com

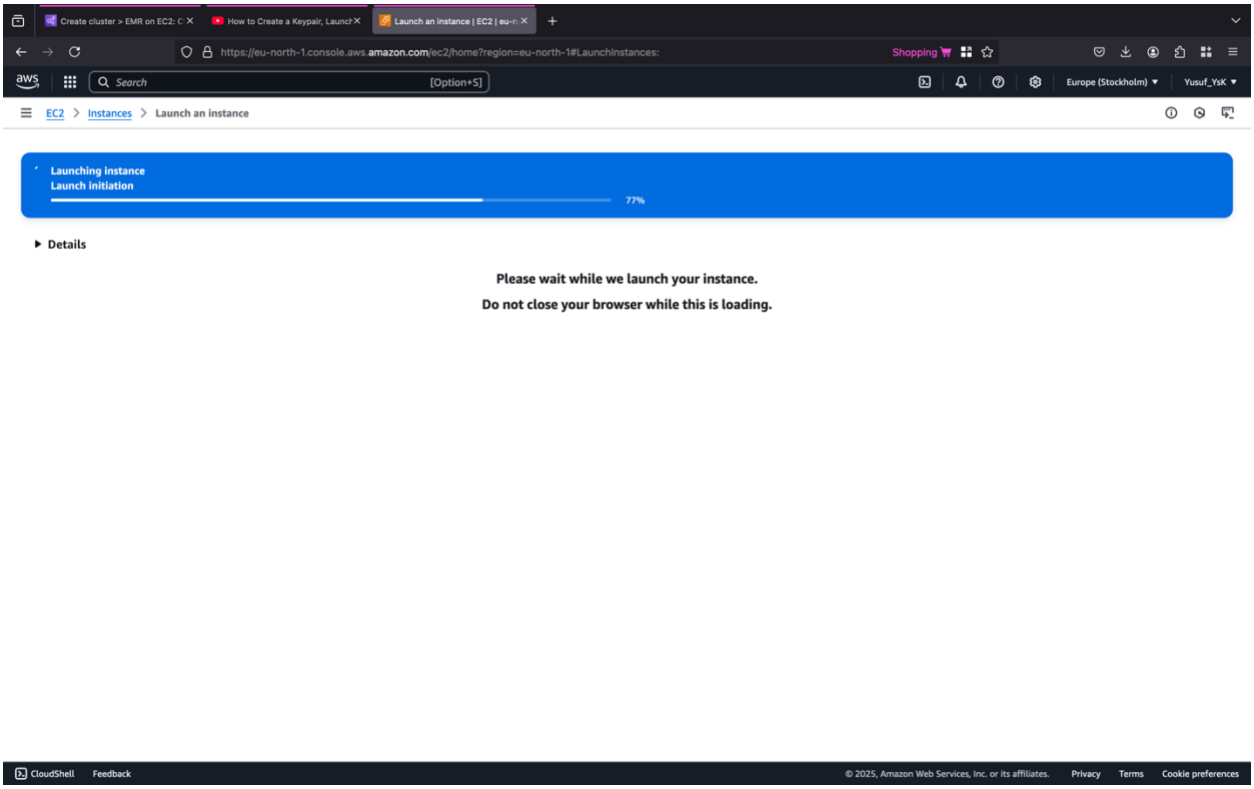


I create a python script with nano for pysaprk emr.

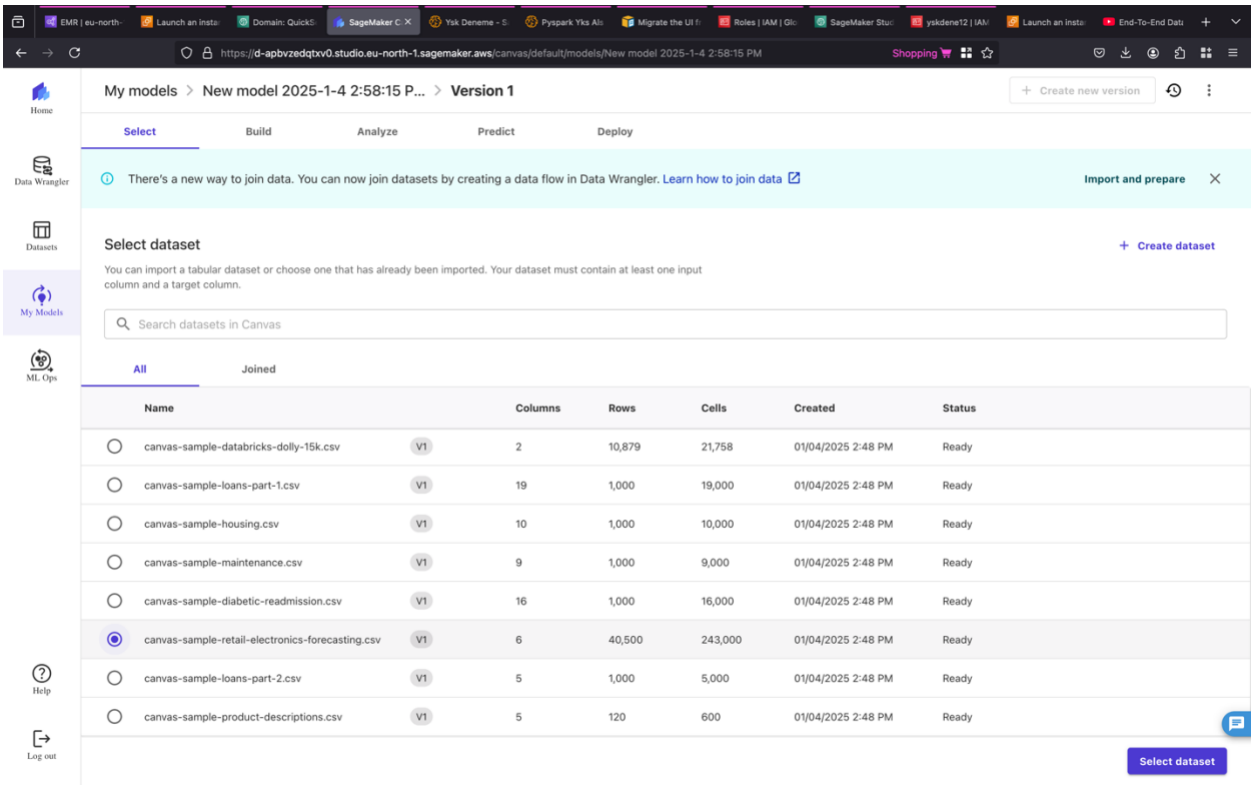


Once the necessary Python code was ready, I submitted it to the EMR cluster for execution. In the submission process, I specified the addresses of the S3 buckets that I had previously created, ensuring that the EMR cluster could access the stored data for processing. By

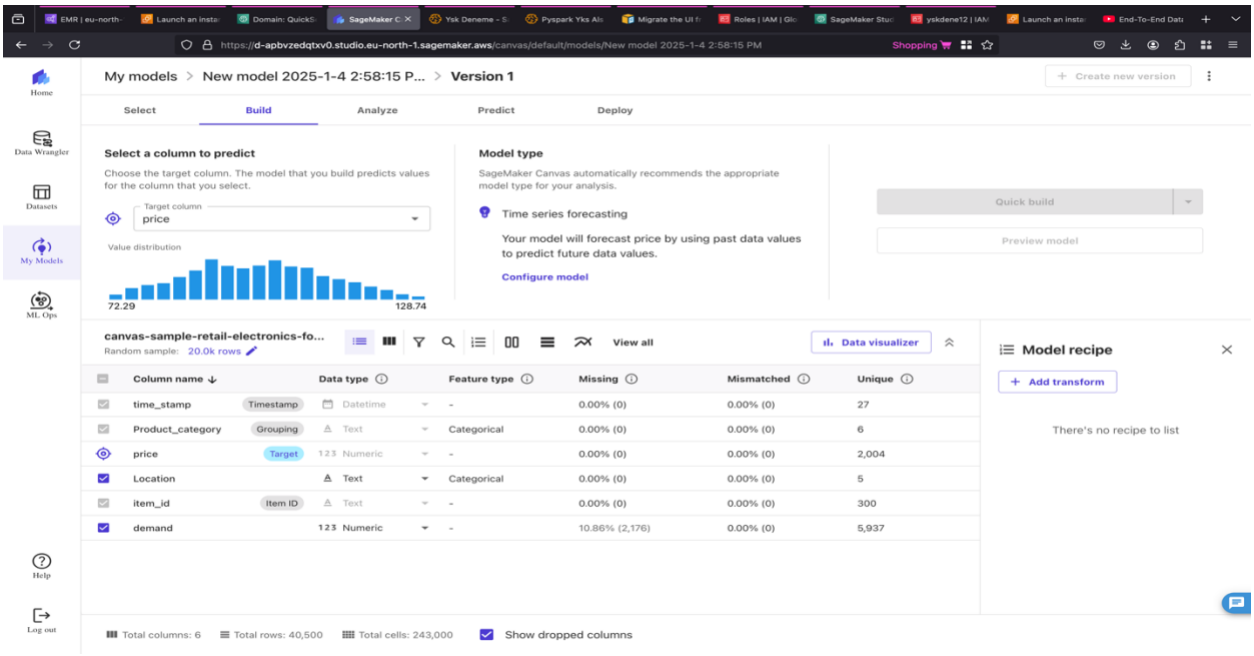
pointing to these S3 locations, the cluster was able to retrieve the dataset and perform the required data processing tasks, such as training the machine learning model.

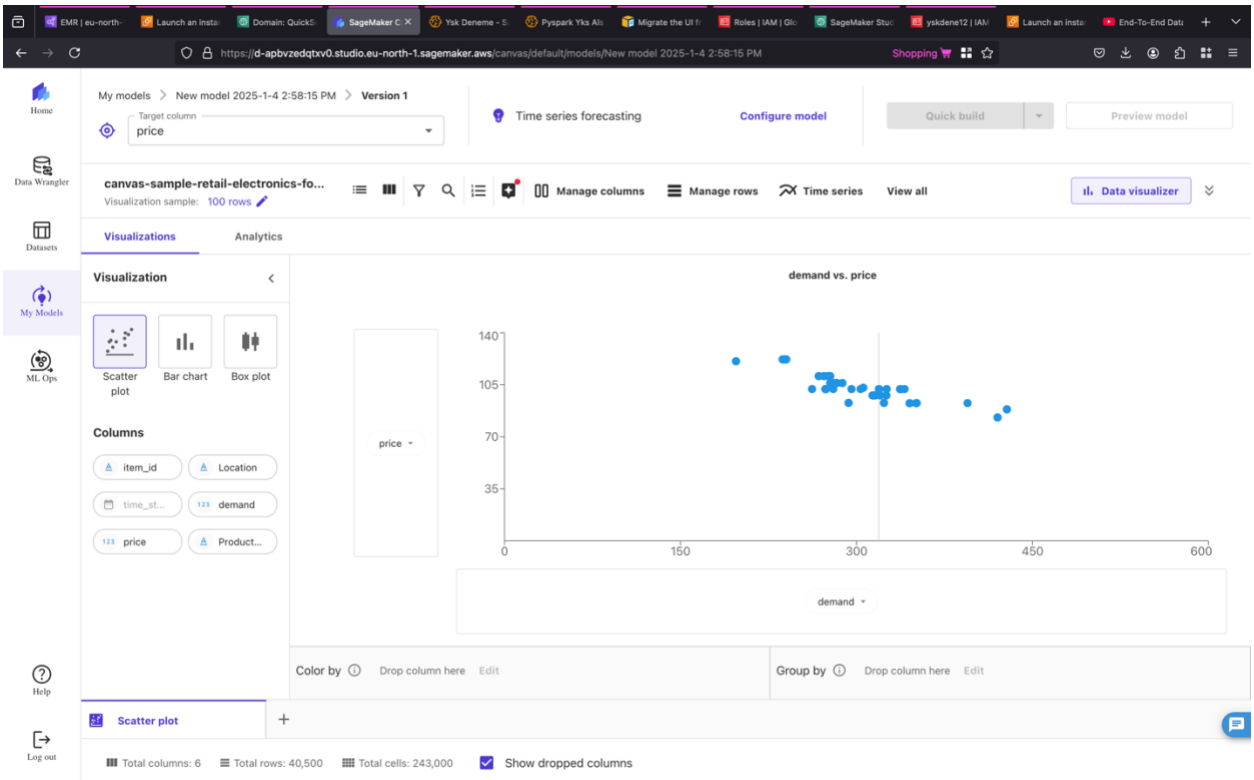


I also worked with EC2 instances as part of my project infrastructure. EC2 (Elastic Compute Cloud) provides scalable computing capacity, which I used to run custom scripts, manage resources, and support various computational tasks that complement the workflows in my AWS environment. By configuring EC2 instances to interact with other AWS services like S3 and EMR, I was able to ensure smooth integration and efficient execution of the machine learning pipeline.

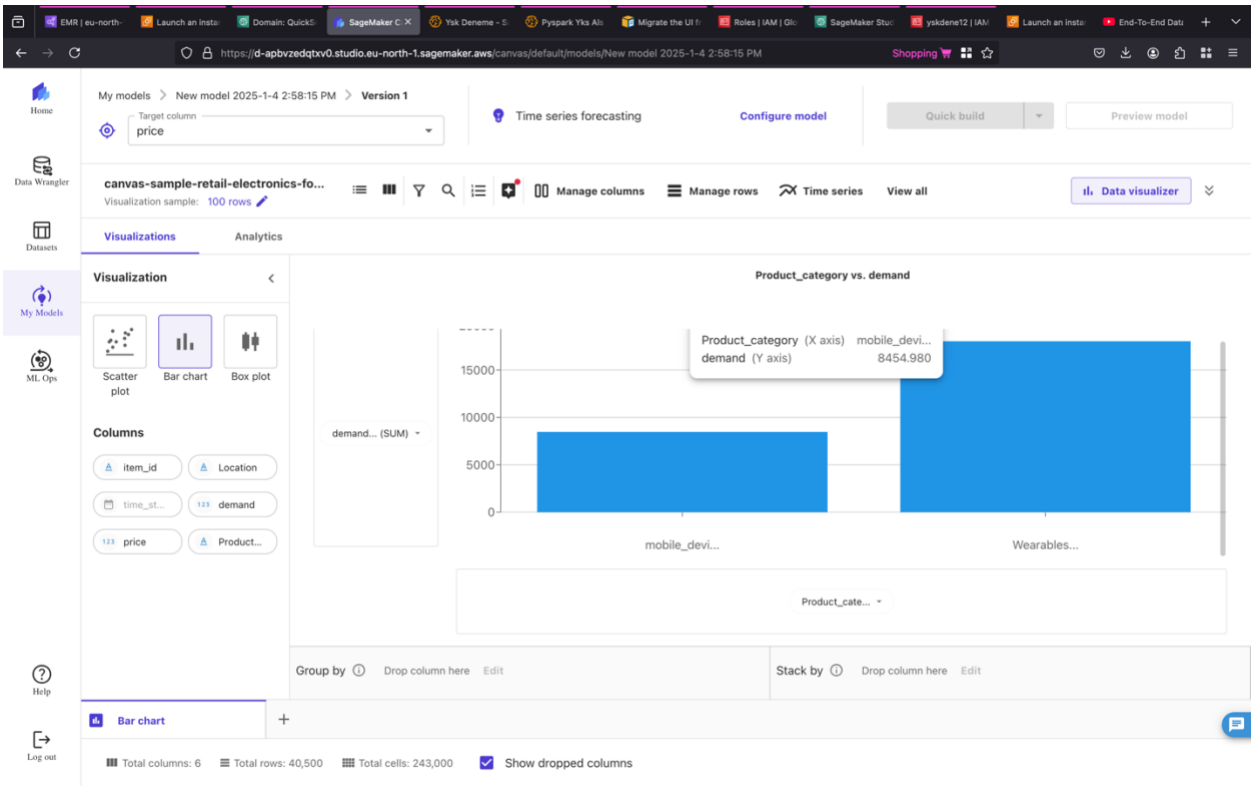


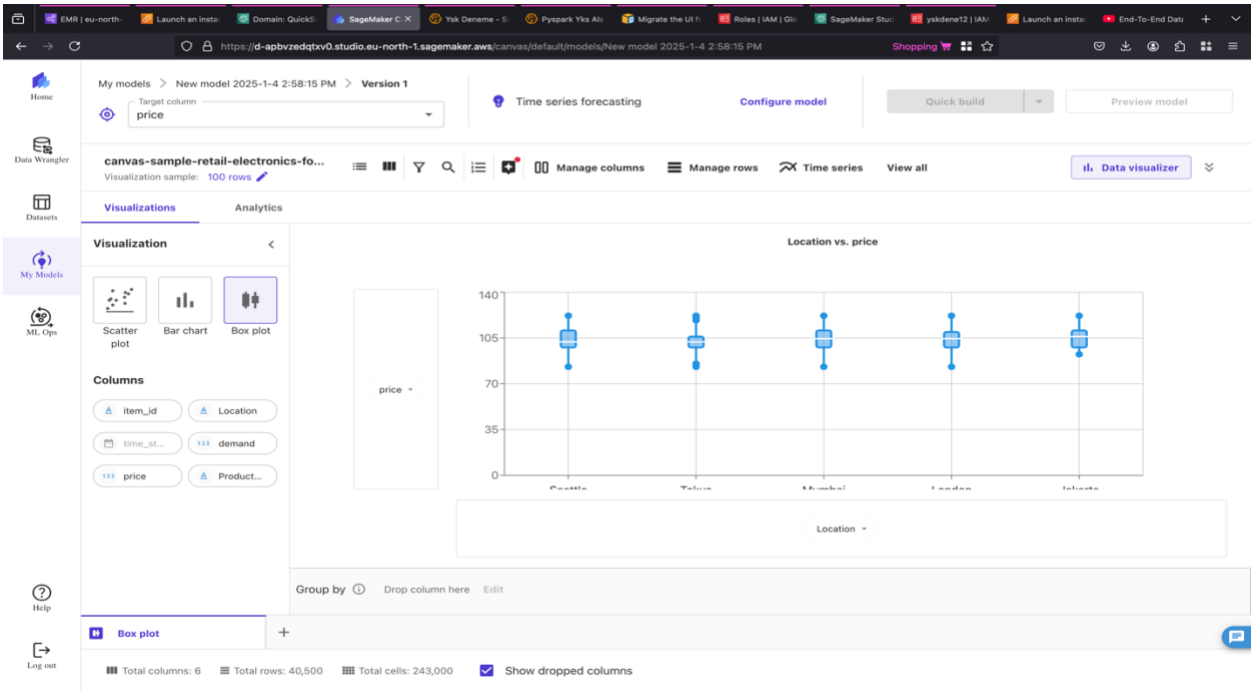
To further expand my knowledge of AWS and explore additional tools, I began using AWS SageMaker Canvas. SageMaker Canvas is a visual, no-code interface that allows users to build and train machine learning models without writing any code. By experimenting with this service, I was able to deepen my understanding of AWS's machine learning offerings and gain hands-on experience with model building, data exploration, and automated predictions. This tool provided an intuitive way to apply machine learning concepts while continuing my learning journey within the AWS ecosystem.

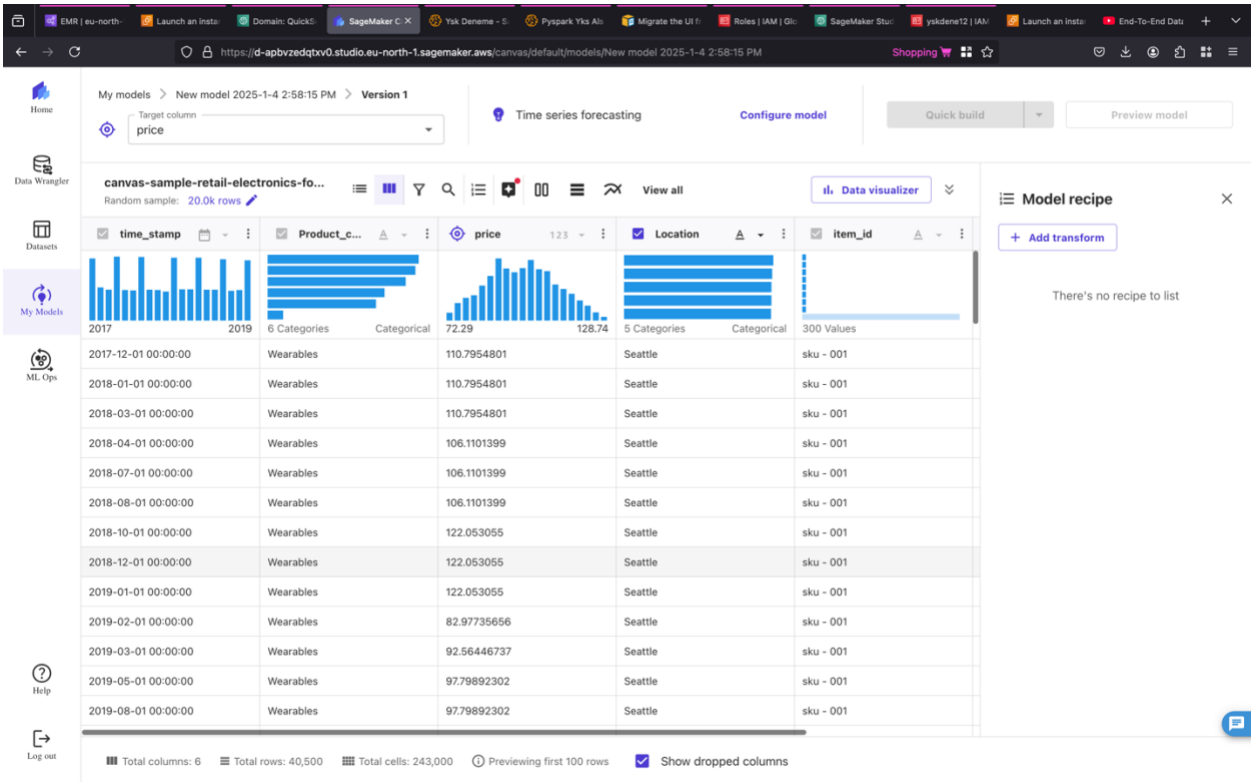

My models > New model 2025-1-4 2:58:15 P... > Version 1
Create new version
Select
Build
Analyze
Predict
Deploy
Select a column to predict
Choose the target column. The model that you build predicts values for the column that you select.
Target column
price
Value distribution
72.29
128.74
Model type
SageMaker Canvas automatically recommends the appropriate model type for your analysis.
Time series forecasting
Your model will forecast price by using past data values to predict future data values.
Configure model
Quick build
Preview model
canvas-sample-retail-electronics-fo...
Random sample: 20.0k rows
View all
Data visualizer
Model recipe
Add transform
There's no recipe to list
Column name
Data type
Feature type
Missing
Mismatched
Unique
time_stamp Timestamp Datetime - 0.00% (0) 0.00% (0) 27
Product_category Grouping Text Categorical 0.00% (0) 0.00% (0) 6
price Target Numeric - 0.00% (0) 0.00% (0) 2,004
Location Text Categorical 0.00% (0) 0.00% (0) 5
item_id Item ID Text - 0.00% (0) 0.00% (0) 300
demand Numeric - 10.86% (2,176) 0.00% (0) 5,937
Help
Log out
Total columns: 6
Total rows: 40,500
Total cells: 243,000
Show dropped columns


My models > New model 2025-1-4 2:58:15 PM > Version 1
Target column
price
Time series forecasting
Configure model
Quick build
Preview model
canvas-sample-retail-electronics-fo...
Visualization sample: 100 rows
Manage columns
Manage rows
Time series
View all
Data visualizer
Visualizations
Analytics
Visualization
Scatter plot
Bar chart
Box plot
Columns
item_id
Location
time_st...
demand
price
Product...
demand vs. price
price
demand
140
105
70
35
0
150
300
450
600
Color by
Drop column here
Edit
Group by
Drop column here
Edit
Scatter plot
Help
Log out
Total columns: 6
Total rows: 40,500
Total cells: 243,000
Show dropped columns

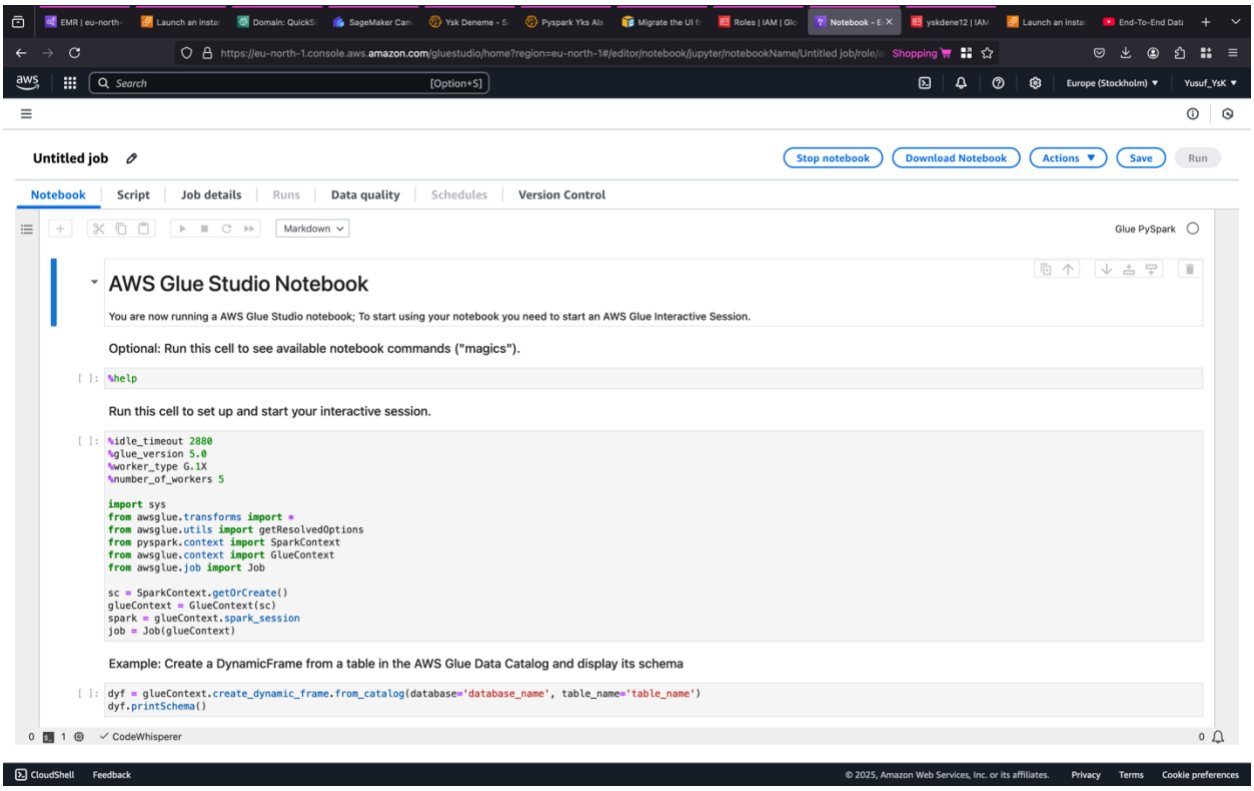
To further explore AWS SageMaker Canvas, I used a pre-existing dataset available within the AWS environment. This allowed me to quickly dive into the machine learning process without needing to prepare or upload a new dataset. I examined the various charts and visualizations generated by the platform, which provided valuable insights into the data's patterns, correlations, and potential trends. By analyzing these visual outputs, I gained a better understanding of how SageMaker Canvas can streamline the data analysis and model-building process.

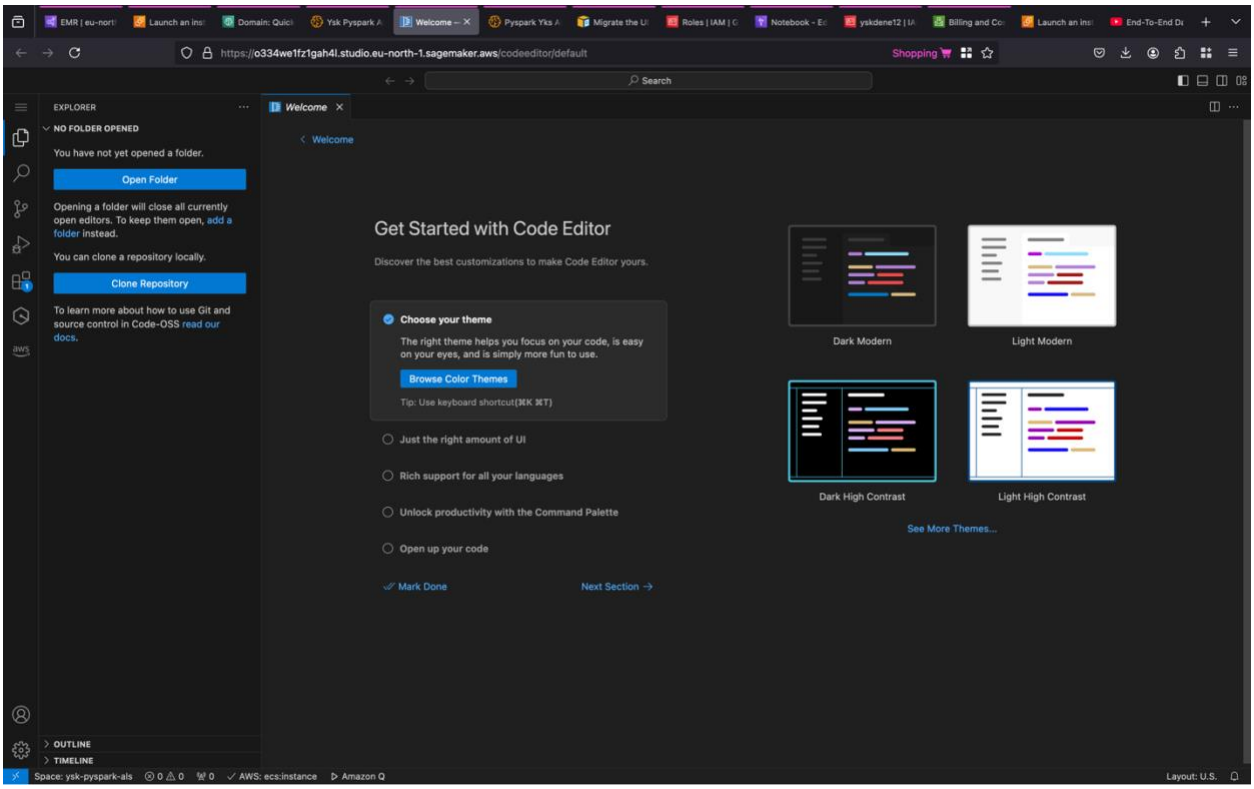



My models > New model 2025-1-4 2:58:15 PM > Version 1
Target column
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Time series forecasting
Configure model
Quick build
Preview model
canvas-sample-retail-electronics-fo...
Visualization sample: 100 rows
Manage columns
Manage rows
Time series
View all
Data visualizer
Visualizations
Analytics
Visualization
Scatter plot
Bar chart
Box plot
Columns
item_id
Location
time_st...
demand
price
Product...
Location vs. price
140
105
70
35
0
price
Location
Group by
Drop column here
Edit
Box plot
Total columns: 6
Total rows: 40,500
Total cells: 243,000
Show dropped columns

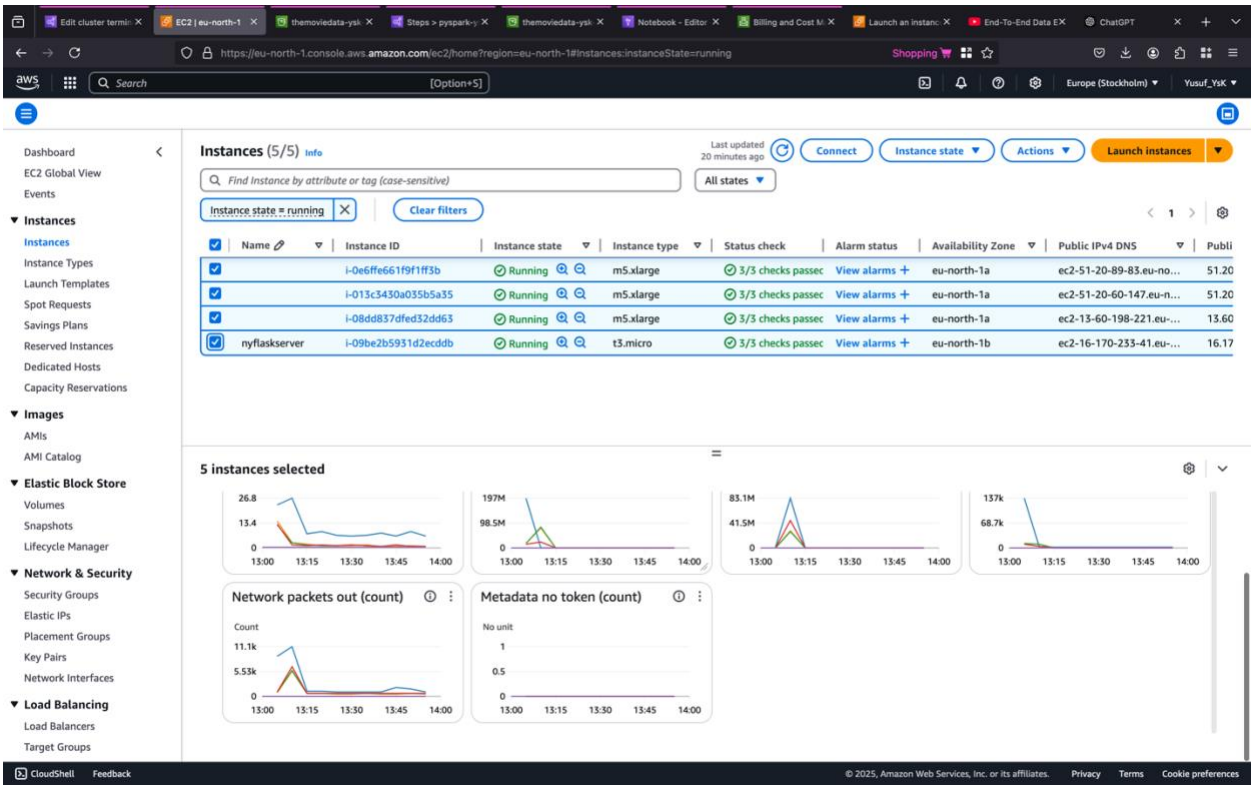

My models > New model 2025-1-4 2:58:15 PM > Version 1
Target column
price
Time series forecasting
Configure model
Quick build
Preview model
canvas-sample-retail-electronics-fo...
Random sample: 20.0k rows
View all
Data visualizer
Model recipe
Add transform
There's no recipe to list
time_stamp
Product_c...
price
Location
item_id
2017
2019
6 Categories
Categorical
72.29
128.74
5 Categories
Categorical
300 Values
2017-12-01 00:00:00 Wearables 110.7954801 Seattle sku - 001
2018-01-01 00:00:00 Wearables 110.7954801 Seattle sku - 001
2018-03-01 00:00:00 Wearables 110.7954801 Seattle sku - 001
2018-04-01 00:00:00 Wearables 106.1101399 Seattle sku - 001
2018-07-01 00:00:00 Wearables 106.1101399 Seattle sku - 001
2018-08-01 00:00:00 Wearables 106.1101399 Seattle sku - 001
2018-10-01 00:00:00 Wearables 122.053055 Seattle sku - 001
2018-12-01 00:00:00 Wearables 122.053055 Seattle sku - 001
2019-01-01 00:00:00 Wearables 122.053055 Seattle sku - 001
2019-02-01 00:00:00 Wearables 82.97735656 Seattle sku - 001
2019-03-01 00:00:00 Wearables 92.56446737 Seattle sku - 001
2019-05-01 00:00:00 Wearables 97.79892302 Seattle sku - 001
2019-08-01 00:00:00 Wearables 97.79892302 Seattle sku - 001
Total columns: 6
Total rows: 40,500
Total cells: 243,000
Previewing first 100 rows
Show dropped columns

After encountering limitations with AWS Learner Lab that prevented me from running scripts previously, I decided to revisit AWS Glue and deepen my understanding by actively working with it. AWS Glue is a fully managed ETL (Extract, Transform, Load) service that facilitates data preparation and integration for analytics. This time, I was able to bypass the restrictions of the Learner Lab environment and fully leverage Glue's capabilities to perform tasks like data cataloging, job scheduling, and transforming data for use in other AWS services. Hands-on experimentation allowed me to apply my learning in a practical context and explore Glue’s role in the overall machine learning pipeline.

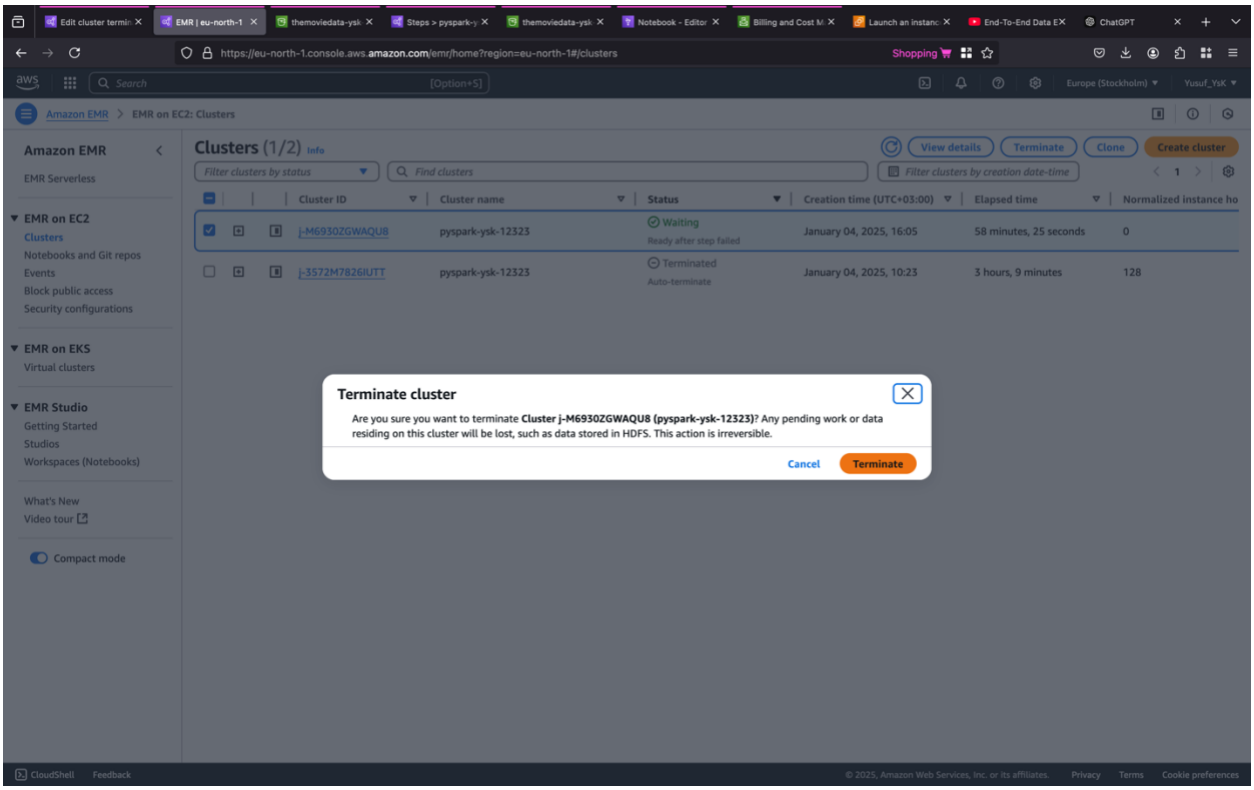




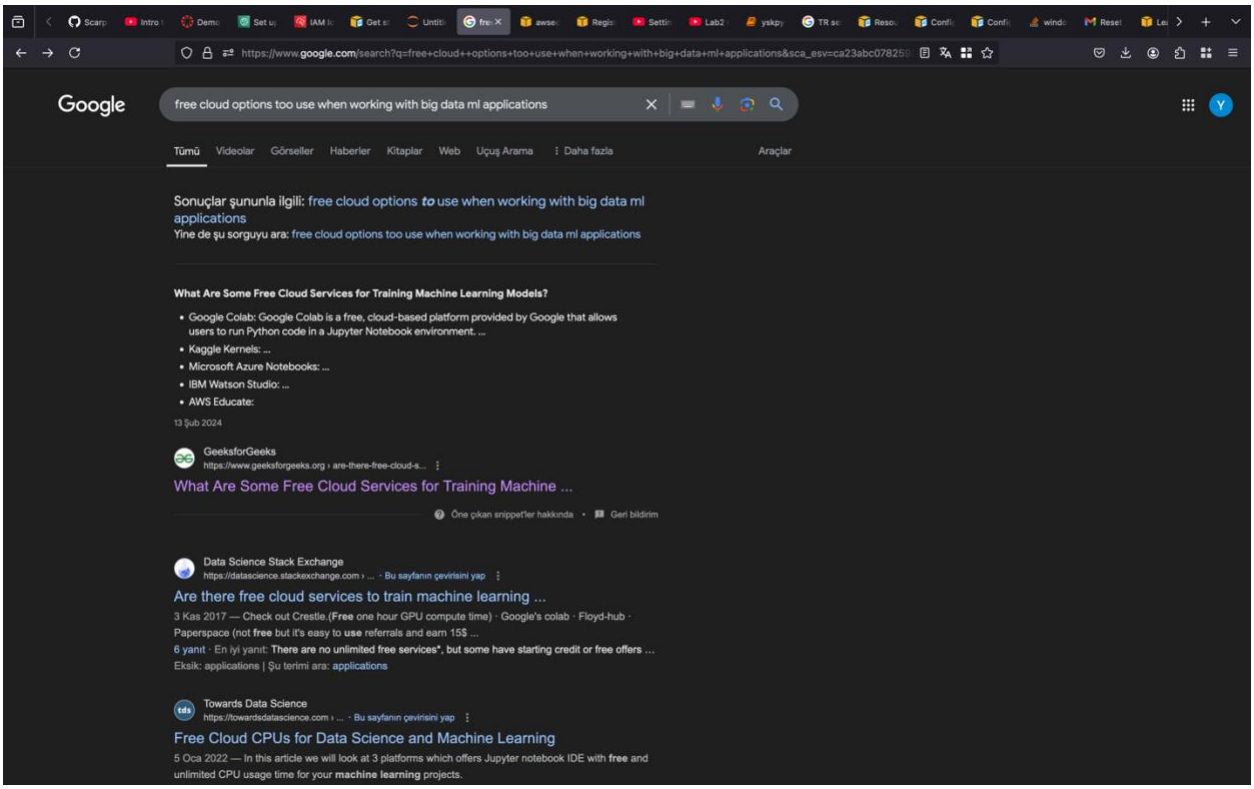
I also explored the SageMaker Code Editor, which provides an integrated development environment (IDE) similar to Visual Studio Code. This cloud-based editor allows users to write, edit, and run code directly within the AWS SageMaker environment. The experience was highly intuitive, with features such as code completion, debugging, and version control support, making it easier to develop machine learning models. By using SageMaker Code Editor, I was able to streamline my coding process and seamlessly interact with other AWS services in a familiar and efficient workspace.



The running instances.



After I am done I terminate everything and look for free options.


https://www.google.com/search?q=free+cloud++options+too+use+when+working+with+big+data+ml+applications&sca_esv=ca23abc078259
Google
free cloud options too use when working with big data ml applications
Tümü
Videolar
Görseller
Haberler
Kitaplar
Web
Uçuş Arama
Daha fazla
Araçlar
Sonuçlar şununla ilgili: free cloud options to use when working with big data ml applications
Yine de şu sorguyu ara: free cloud options too use when working with big data ml applications
What Are Some Free Cloud Services for Training Machine Learning Models?
Google Colab: Google Colab is a free, cloud-based platform provided by Google that allows users to run Python code in a Jupyter Notebook environment. ...
Kaggle Kernels: ...
Microsoft Azure Notebooks: ...
IBM Watson Studio: ...
AWS Educate:
13 Şub 2024
GeeksforGeeks
https://www.geeksforgeeks.org › are-there-free-cloud-s...
What Are Some Free Cloud Services for Training Machine ...
Öne çıkan snippet'ler hakkında
Geri bildirim
Data Science Stack Exchange
https://datascience.stackexchange.com › ...
Bu sayfanın çevirisini yap
Are there free cloud services to train machine learning ...
3 Kas 2017 — Check out Crestle.(Free one hour GPU compute time) · Google's colab · Floyd-hub · Paperspace (not free but it's easy to use referrals and earn 15$ ...
6 yanıt · En iyi yanıt: There are no unlimited free services*, but some have starting credit or free offers ...
Eksik: applications | Şu terimi ara: applications
Towards Data Science
https://towardsdatascience.com › ...
Bu sayfanın çevirisini yap
Free Cloud CPUs for Data Science and Machine Learning
5 Oca 2022 — In this article we will look at 3 platforms which offers Jupyter notebook IDE with free and unlimited CPU usage time for your machine learning projects.